FDR Team 4

Junior Design

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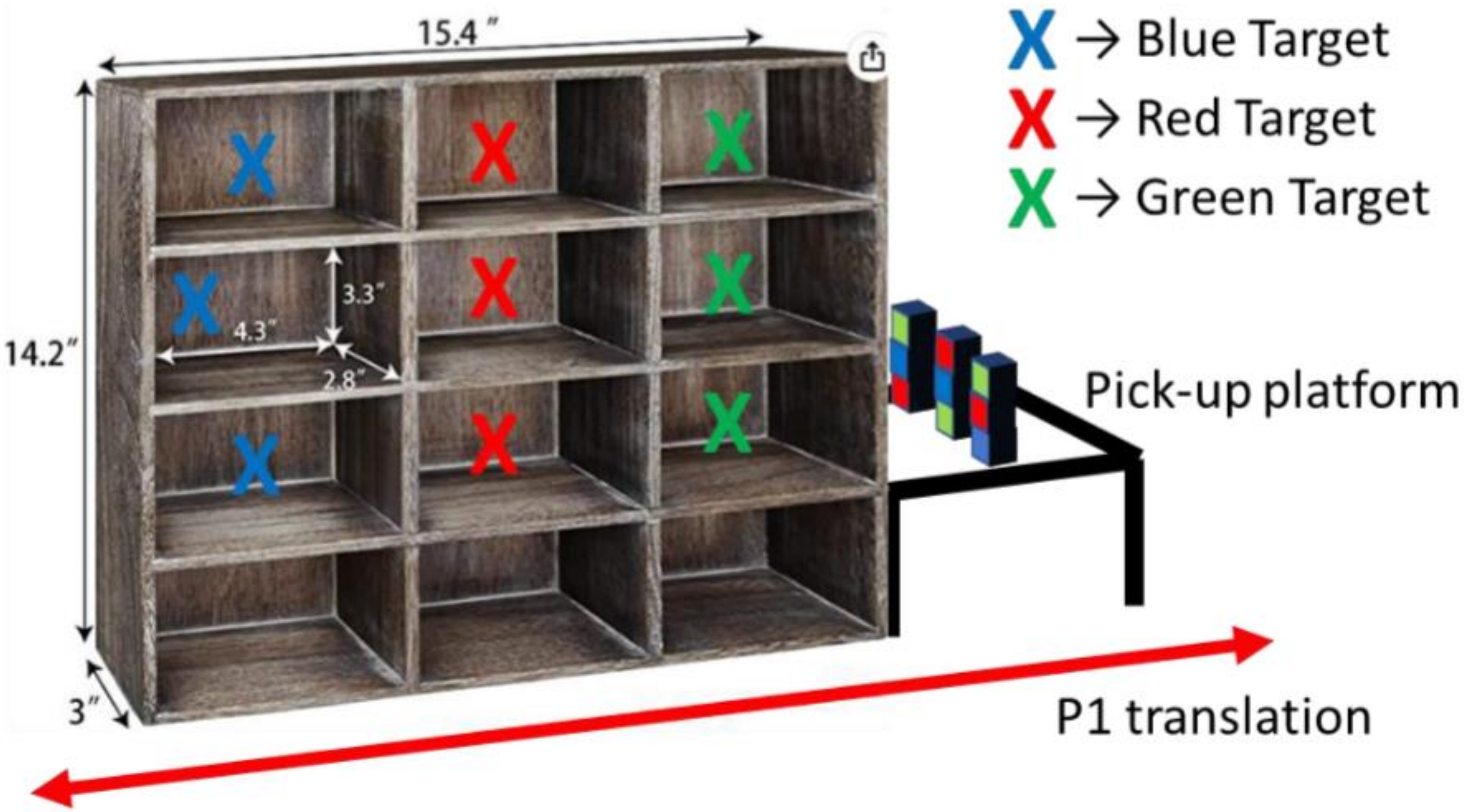
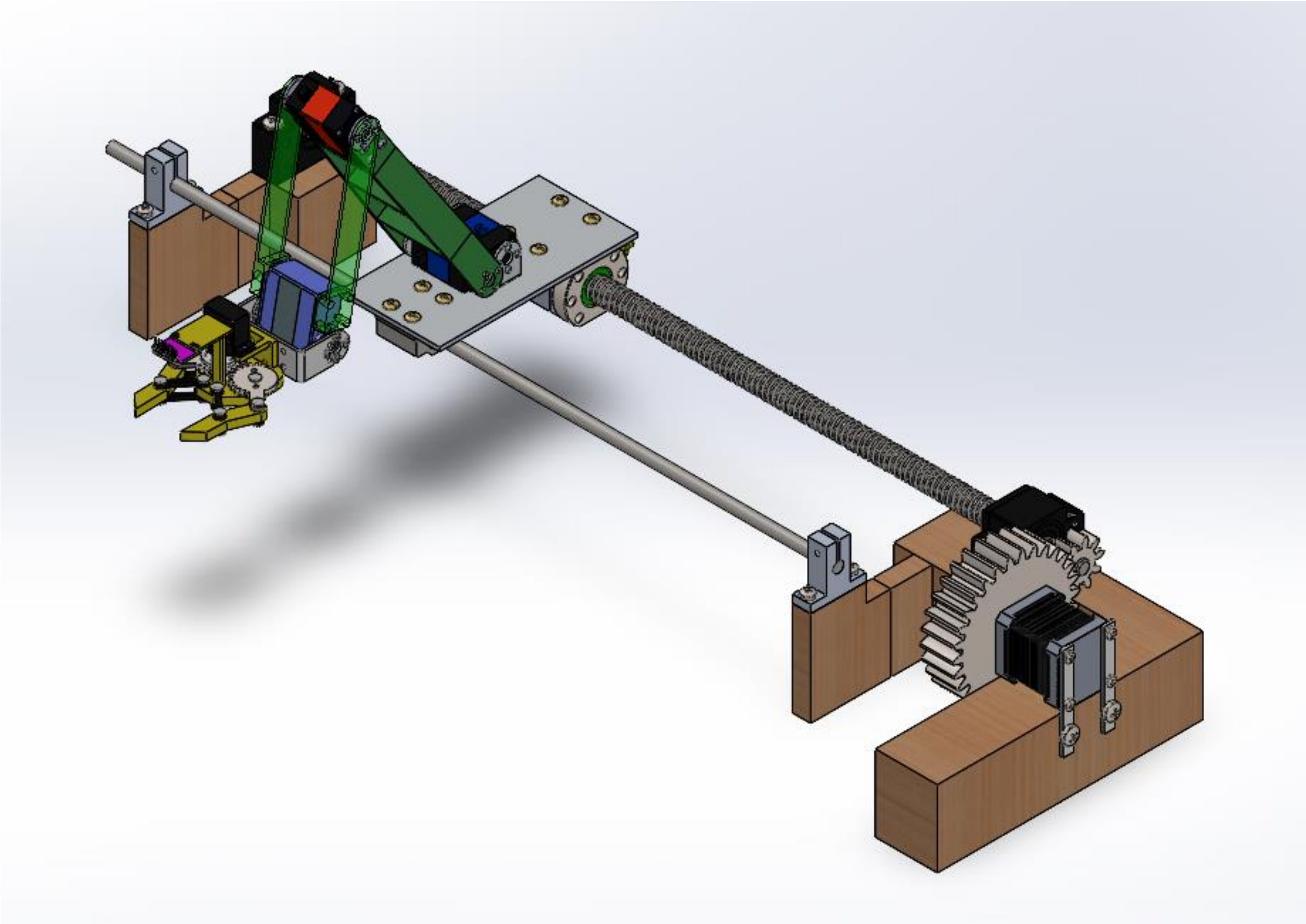
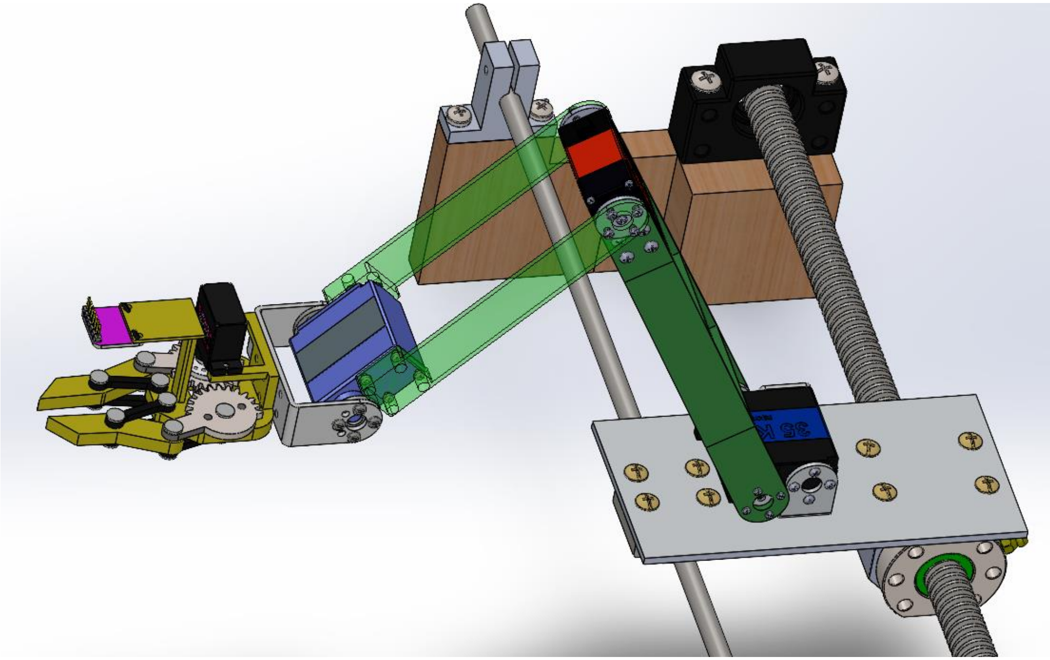
Project Overview



Project Overview

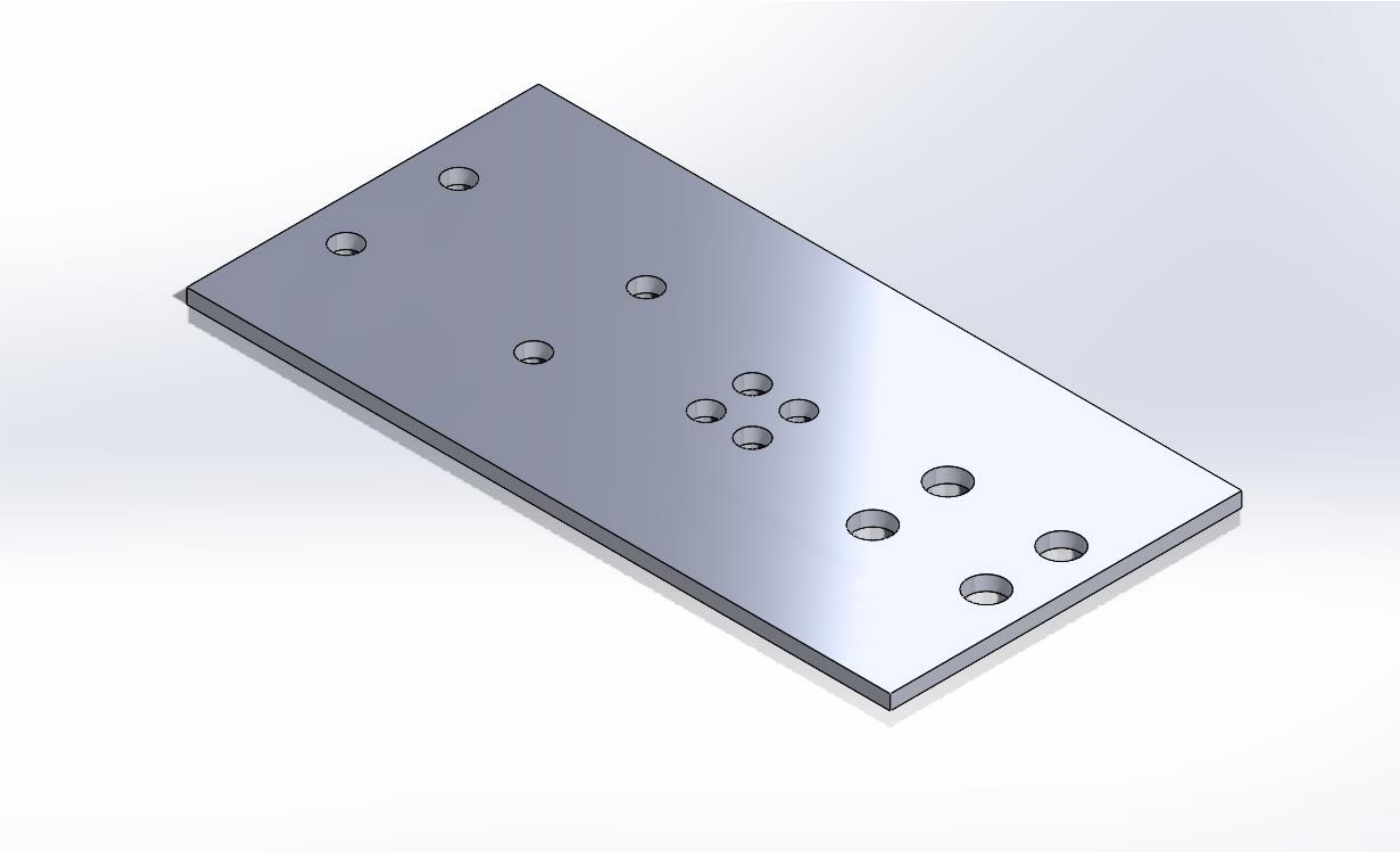




Robot Overview

- Base
- Linear Axis
- Rotary Axis 1
- Rotary Axis 2
- Gripper

Base Configuration

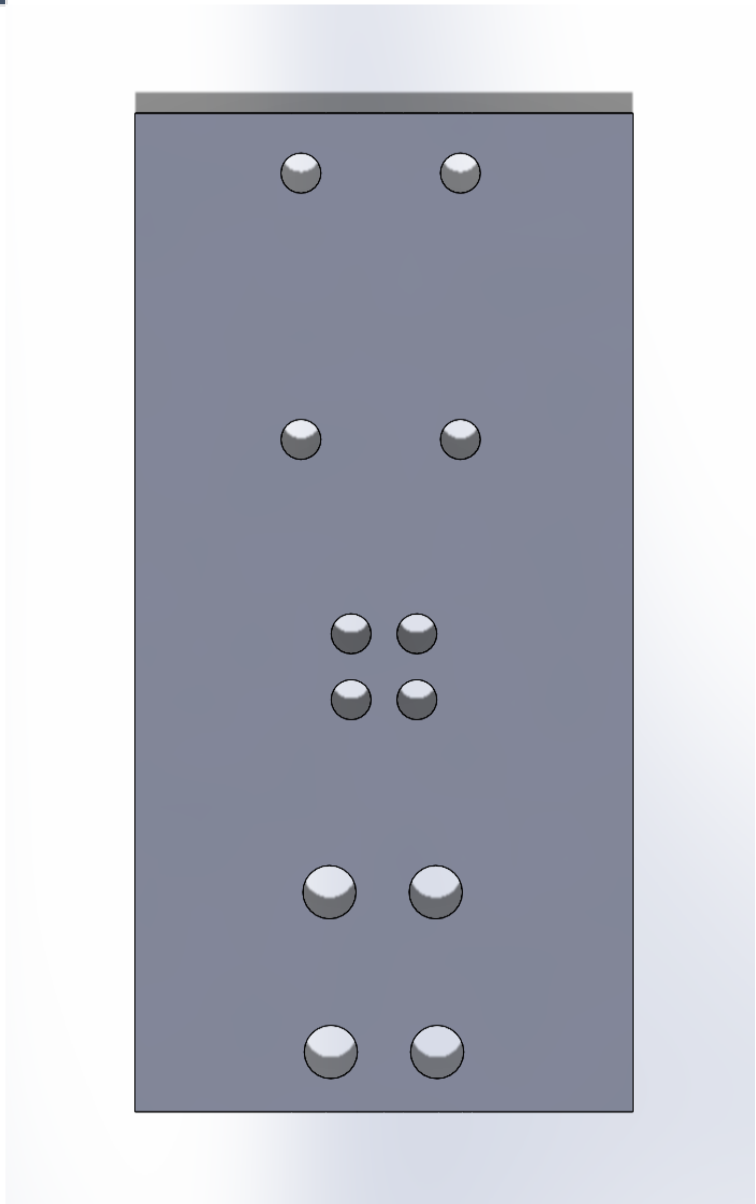


Base Mass

Object	Quantity
Base	1
M3 Holes	8
M4 Holes	4

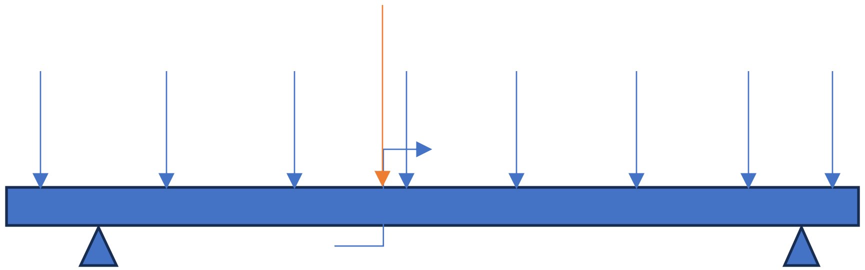
$$Mass = 1(lwt) - 8(\pi r_{M3}^2 t) - 4(\pi r_{M4}^2 t)$$

$$Mass = 136.49\ grams$$



Base Deflection

Variable	Description	Value
W	Distributed load	4.08 N/m
P	Point load	2.744 N
d	right end of beam to right pin	0.021 m
E	Modulus of elacticity (6061 Al)	69 GPa
l	Distance between pins	0.185 m
L	Total length of beam	0.235 m
c	Distance from left end of beam to left pin	0.029 m
a	Distance from left pin to point load	0.054 m
b	Distance from right pin to point load	0.062 m



Marker	Value
Distributed load (blue)	4.08 N/m
Point load (orange)	2.744 N
Moment (blue)	-43.05 Nm

```
deflect =

    1.2168e-18
```

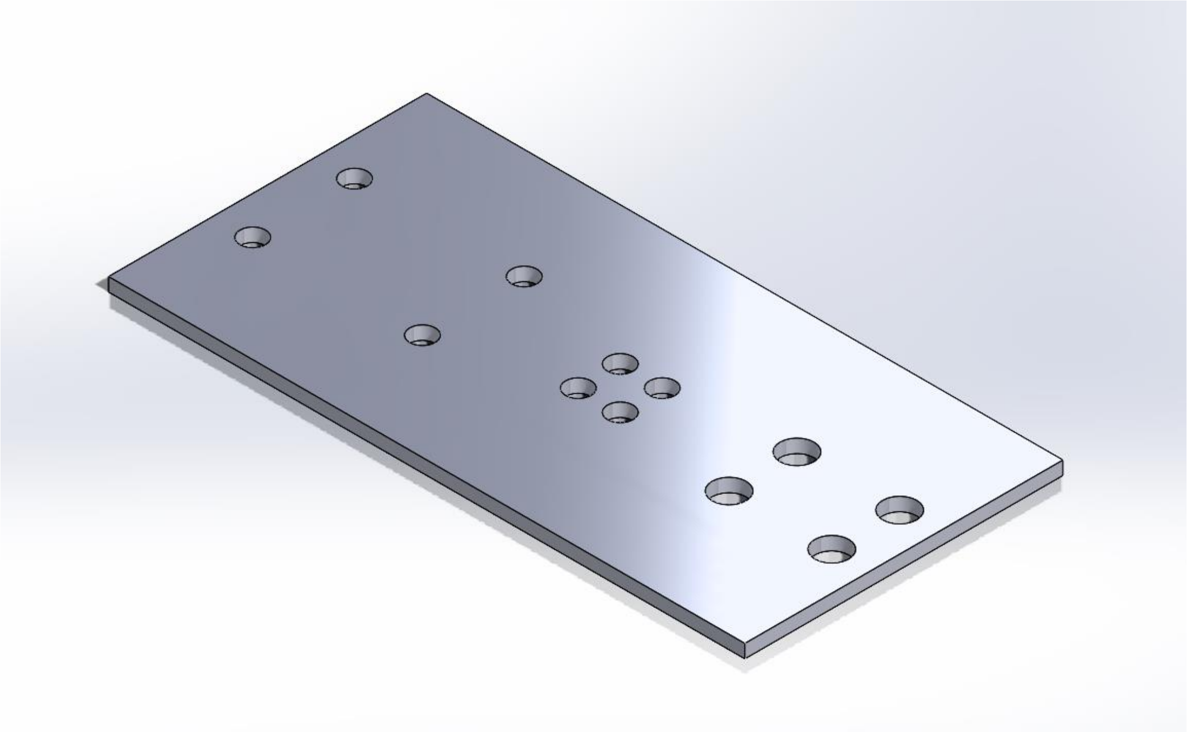
Base Safety Factor

Variable	Value
Von mises stress	0.3766 MPa
Yield strength (Al)	276 MPa

$$SF = \frac{Yield\ Stress\ (Al\ 6061)}{Von\ Mises\ Stress}$$

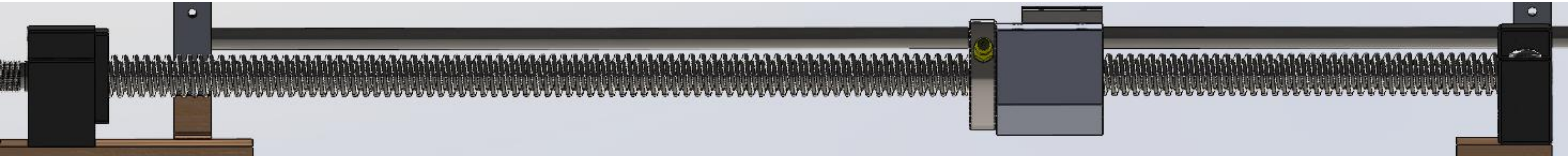
```
sf =

   732.8282
```



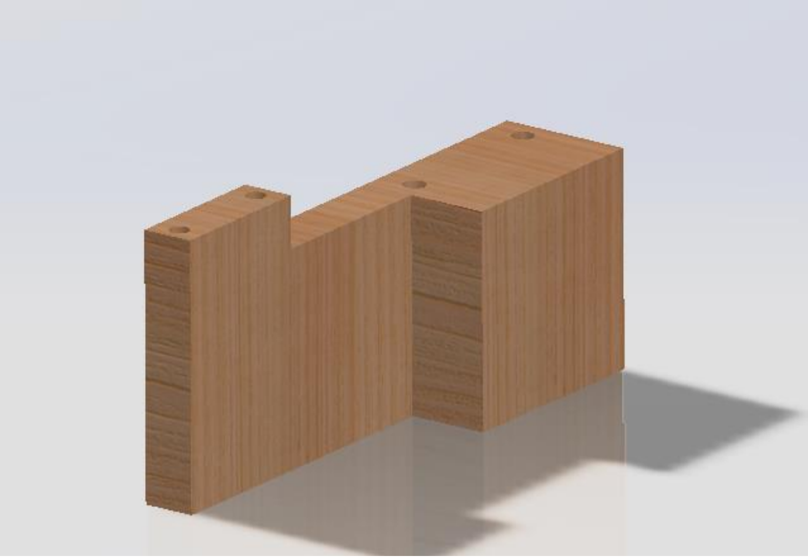
Linear Axis: Configuration Part 1

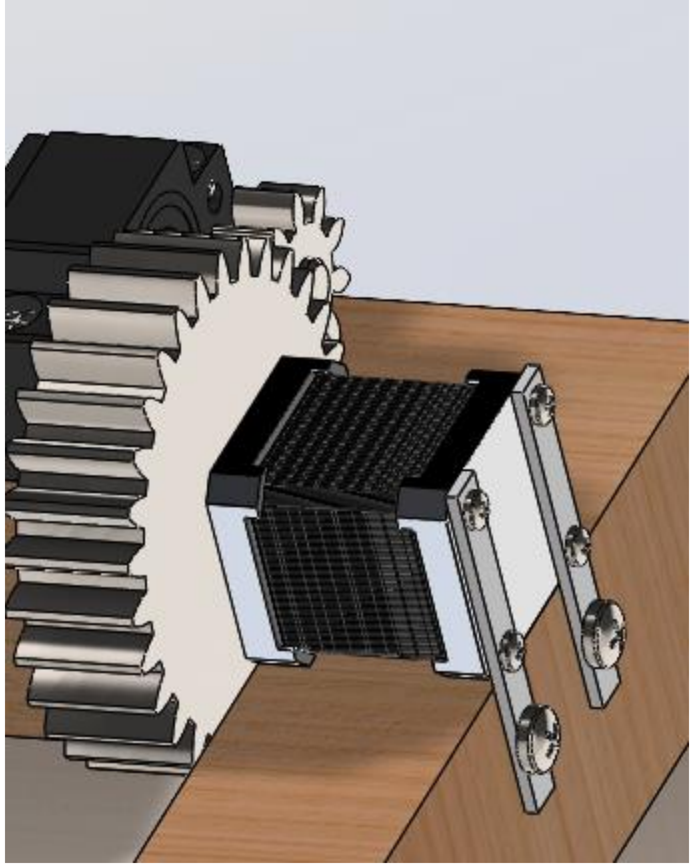
Ball Screw Configuration	Decription
Lead Screw	600mm SFU1605- Bought
Screw Stands	Bk/Bf12- Bought
Ball screw Nut and housing	SFU1605-3 - Bought



Linear Axis: Configuration Part 2

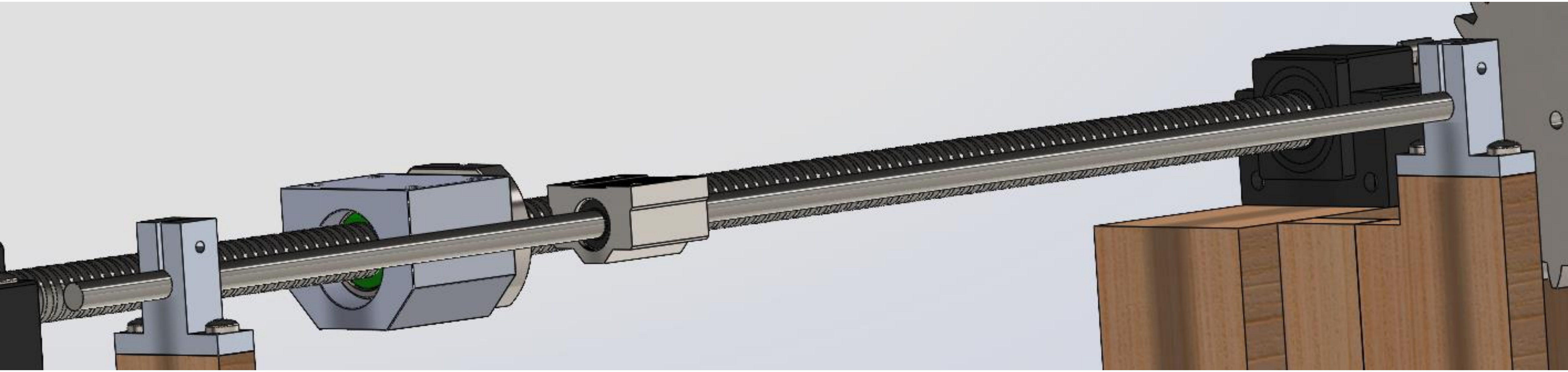
Stepper Motor Assembly	Description
Stepper Motor	NEMA 17, 1.5A, .40Nm -Bought
Gears	Module 3, 30 Teeth and 10 Teeth
Mounting Brackets	Manufactured- 3D Printing
Base	Manufactured- 3D Printing



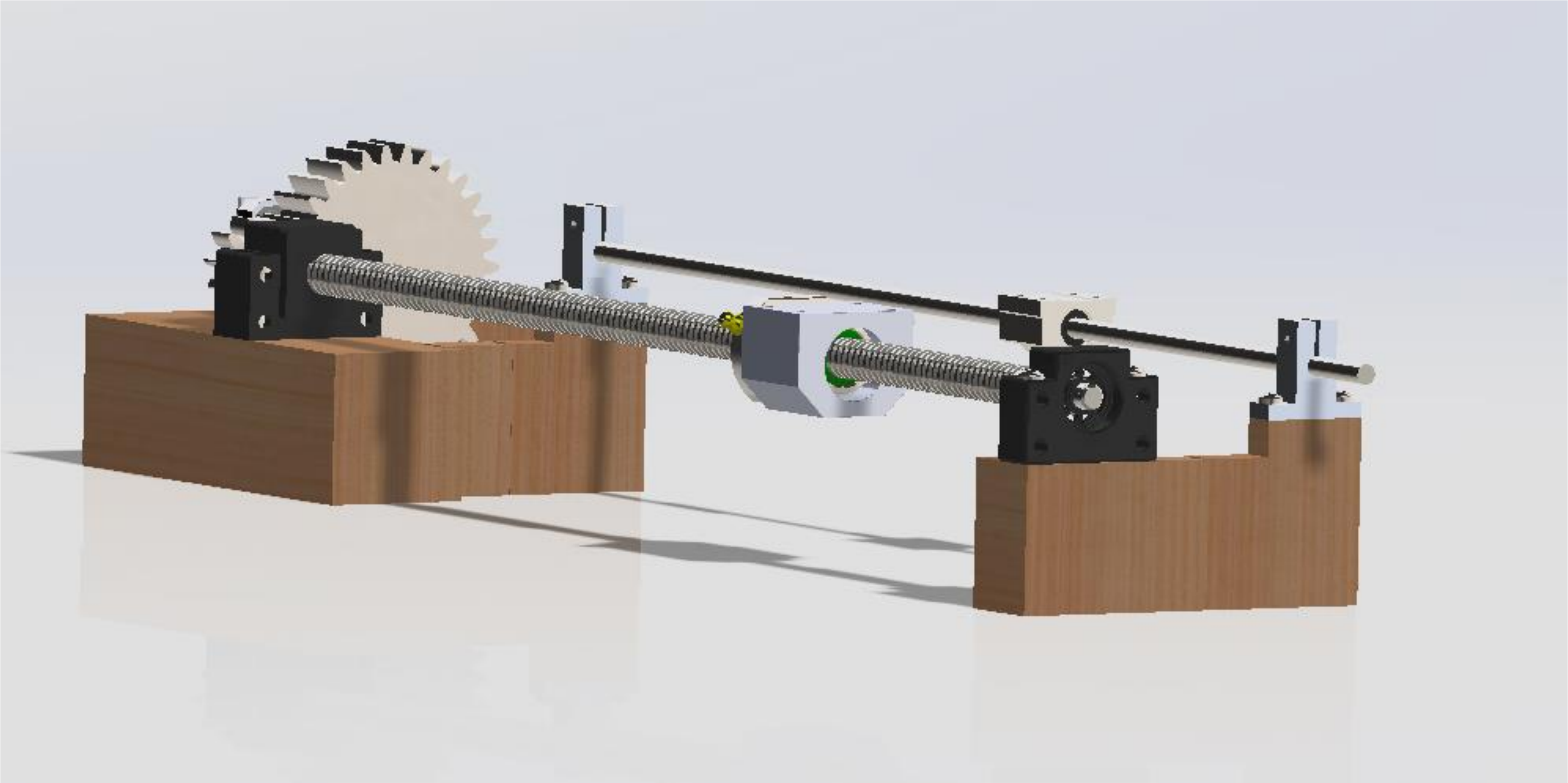


Linear Axis: Configuration Part 3

Guide Rail Assembly	Description
Guide Rails	600mm, Linear Motion Shaft -Bought
Shaft Supports	SK8 Shaft Supports -Bought
Bearings	SCS8UU Bearings -Bought

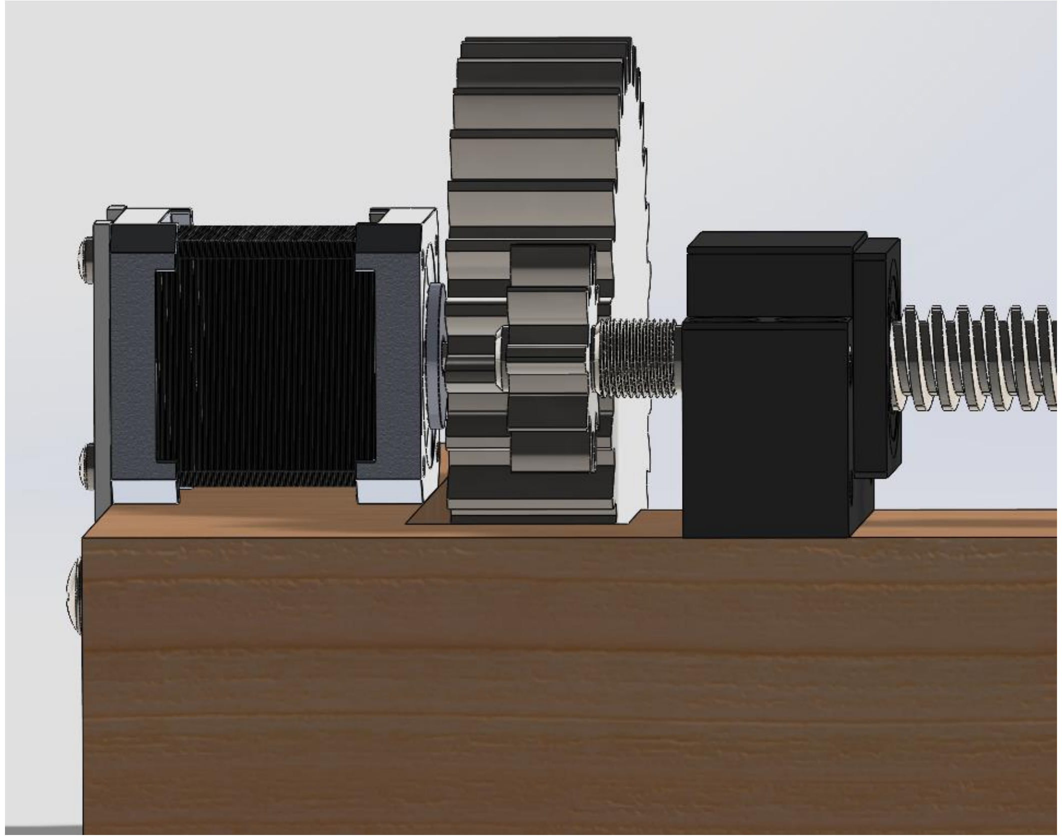


Linear Axis: Final Picture



Linear Axis: Torque and Speed

Variable	Value
Torque Required	.036Nm
Torque Supplied @900rpm	.10Nm
Speed	.225m/s
Acceleration	28m/s^2



$$Safety\ Factor$$

$$SF_1 = \frac{Torque\ of\ Motor}{Torque\ Need} = \frac{.10Nm}{.036Nm} = 2.78$$

$$SF_2 = \frac{Positioning\ Torque}{Torque\ Needed} = \frac{.045Nm}{.0027Nm} = 1.25$$

Linear Axis: Resolution, Mass,and Microswitch

Mass

.448kg

Resolution

Nema 17 Stepper Motor = 1.8°/step

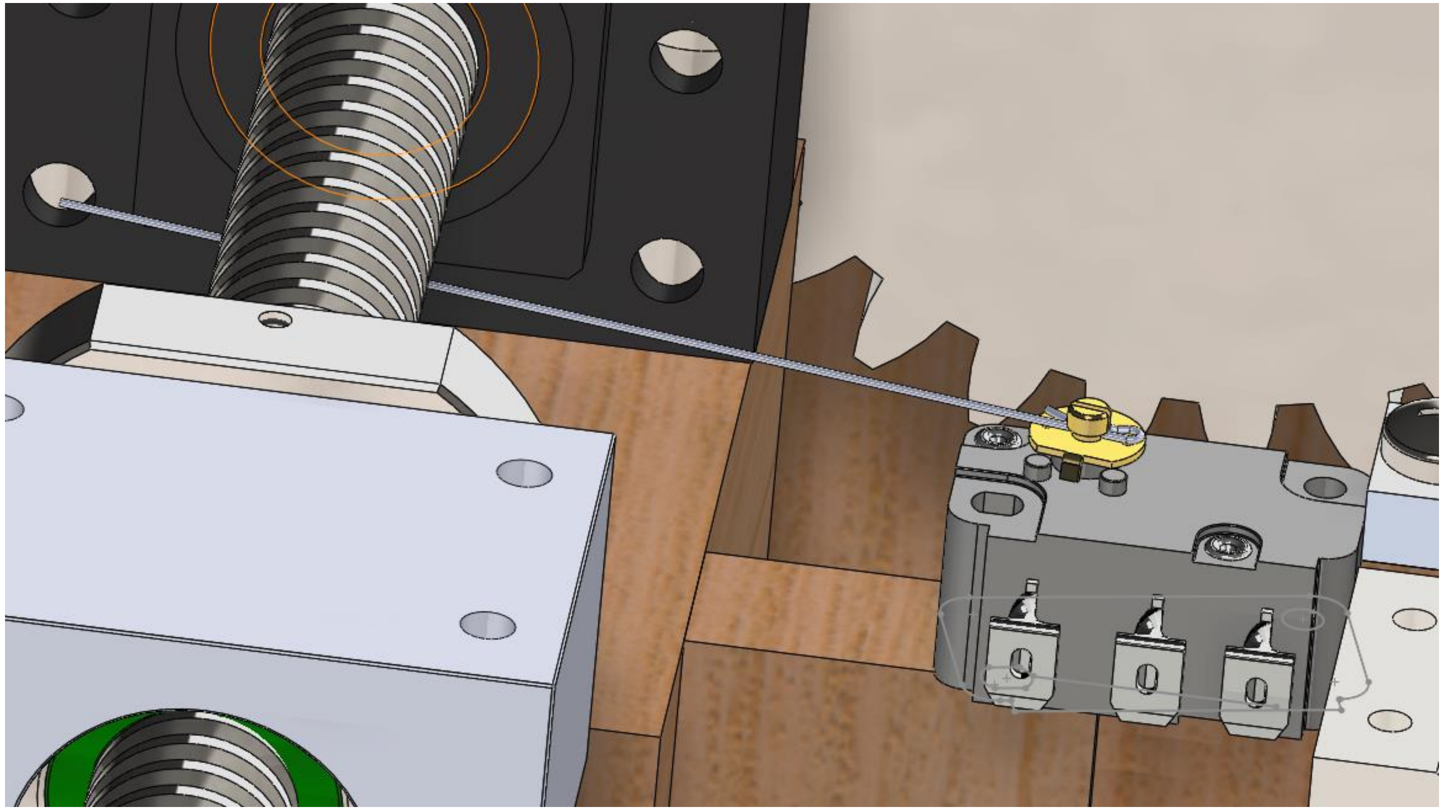
Ball screw Lead = 5mm/rev

200 steps/rev

(5 mm/200 step) = (X mm/ 1 step)

0.025mm/step = Resolution

Microswitch With Wire Trigger



Rotary Axis 1: Configuration

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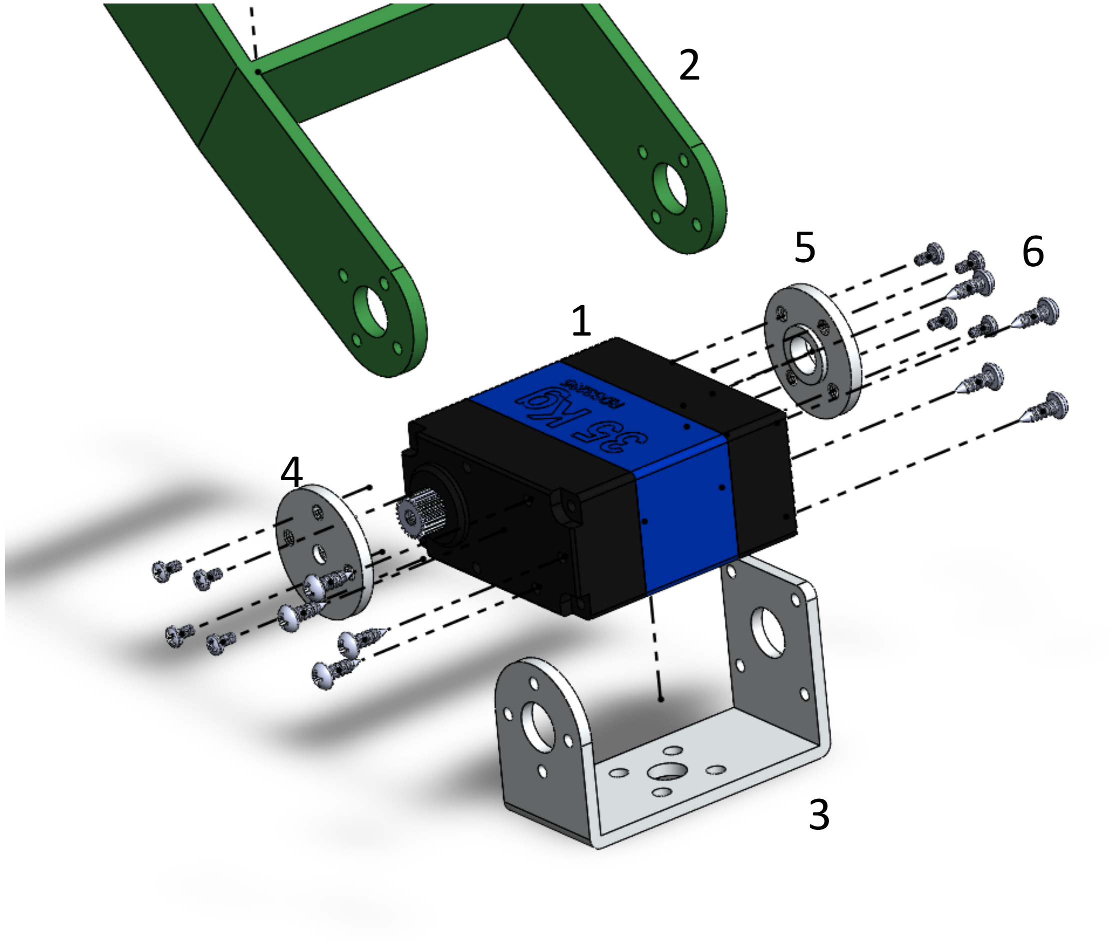


No.	Component	Qty.
1	Double Bearing Servo Motor (35 kg)	1
2	3D Printed Custom Link 1	1
3	Aluminum U-Bracket Mount	1
4	Aluminum Gear Mounted Arm Plate	1
5	Aluminum Arm Plate	1
6	Screws (Included with Servo)	13

Rotary Axis 1: Mass and Safety Factor

Component	Mass	Quantity	Total
Servo Motor Assembly (Motor, Arm Plates, U-Bracket, and Screws)	80 g	1	80 g
Link 1	21.5 g	1	21.5g
		Total	101.5 g

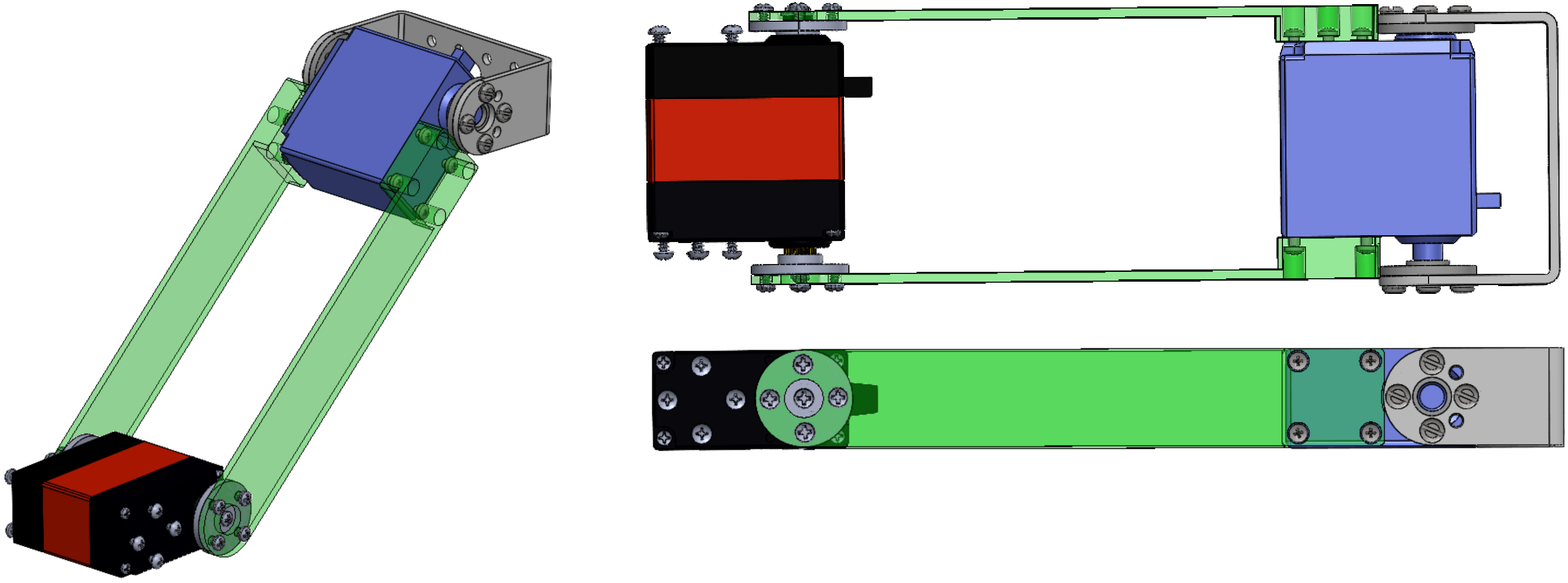
Calculated Max Bending Stress	1.29 MPa
Factor of Safety ($\sigma_y = 26MPa$)	20

Rotary Axis 1: Actuator, Resolution, and Torque

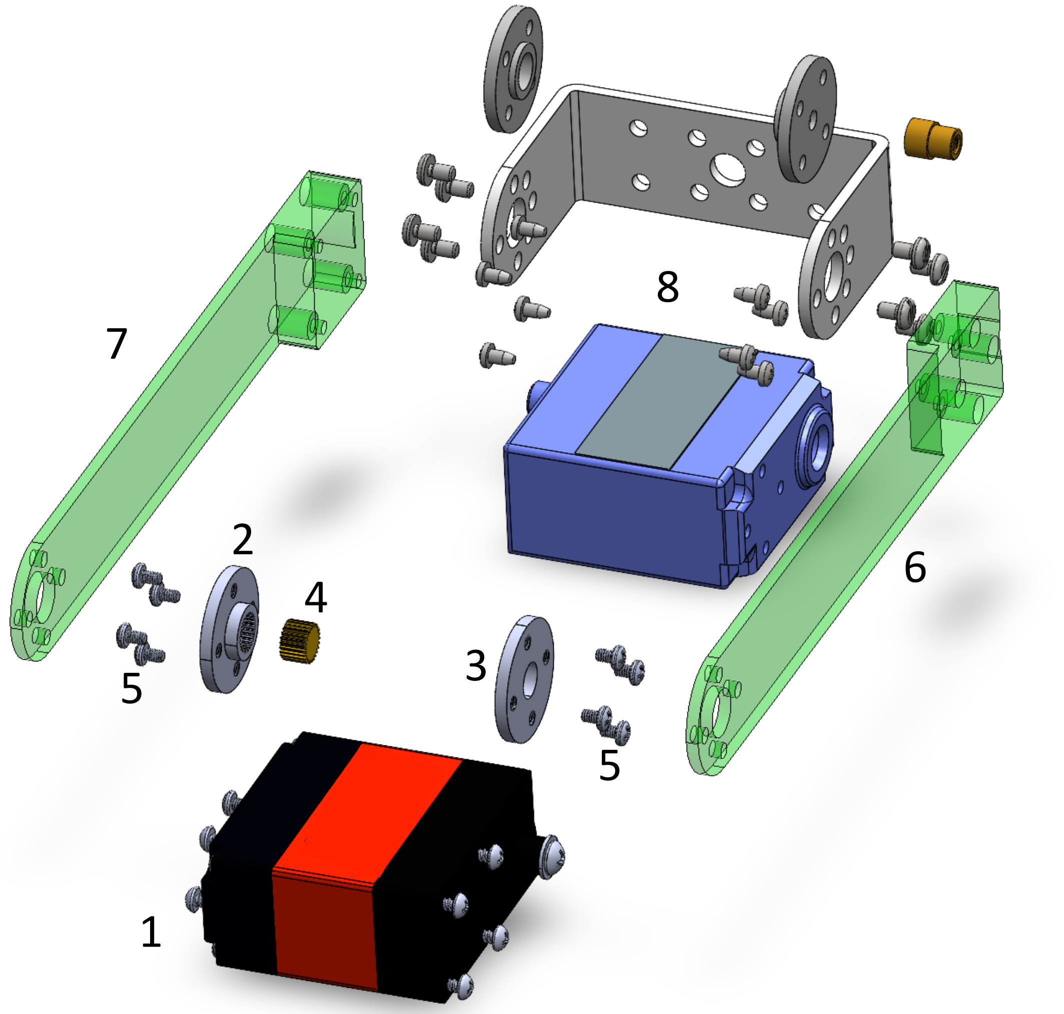
Selected Actuator: RDS3235 Servo

Resolution at End Effector	0.25 cm
Minimum Required Torque	100 Ncm
Torque Provided by Actuator	343 Ncm
Torque Safety Factor	3.4

Rotary Axis 2: Configuration



Rotary Axis 2 Component Sheet



No.	Component	Qty.
1	Double Bearing Servo Motor (20 kg)	1
2	Gear Side Aluminum Mounting Disk*	1
3	Aluminum Mounting Disk*	1
4	Servo Motor Direct Drive Gear*	1
5	Mounting Screws for Link Arms*	8
6	Right Side Link Arm	1
7	Left Side Link Arm	1
8	Gripper Actuation Servo **	1

**These components come with RA 2 Servo*

***This is not included in RA 2 from an analytical standpoint, just a visual aid and for ease when modeling*

Rotary Axis 2: Mass and Safety Factor

Component	Mass	Quantity	Total
Servo Motor Assembly (with all included components)	60 grams	1	60 grams
Left Link 2	6.05 grams	1	6.05 grams
Right Link 2	6.05 grams	1	6.05 grams
		Total	72.1 grams

Calculated Max Bending Stress	0.5390 MPa
Factor of Safety (FS)	48.2349

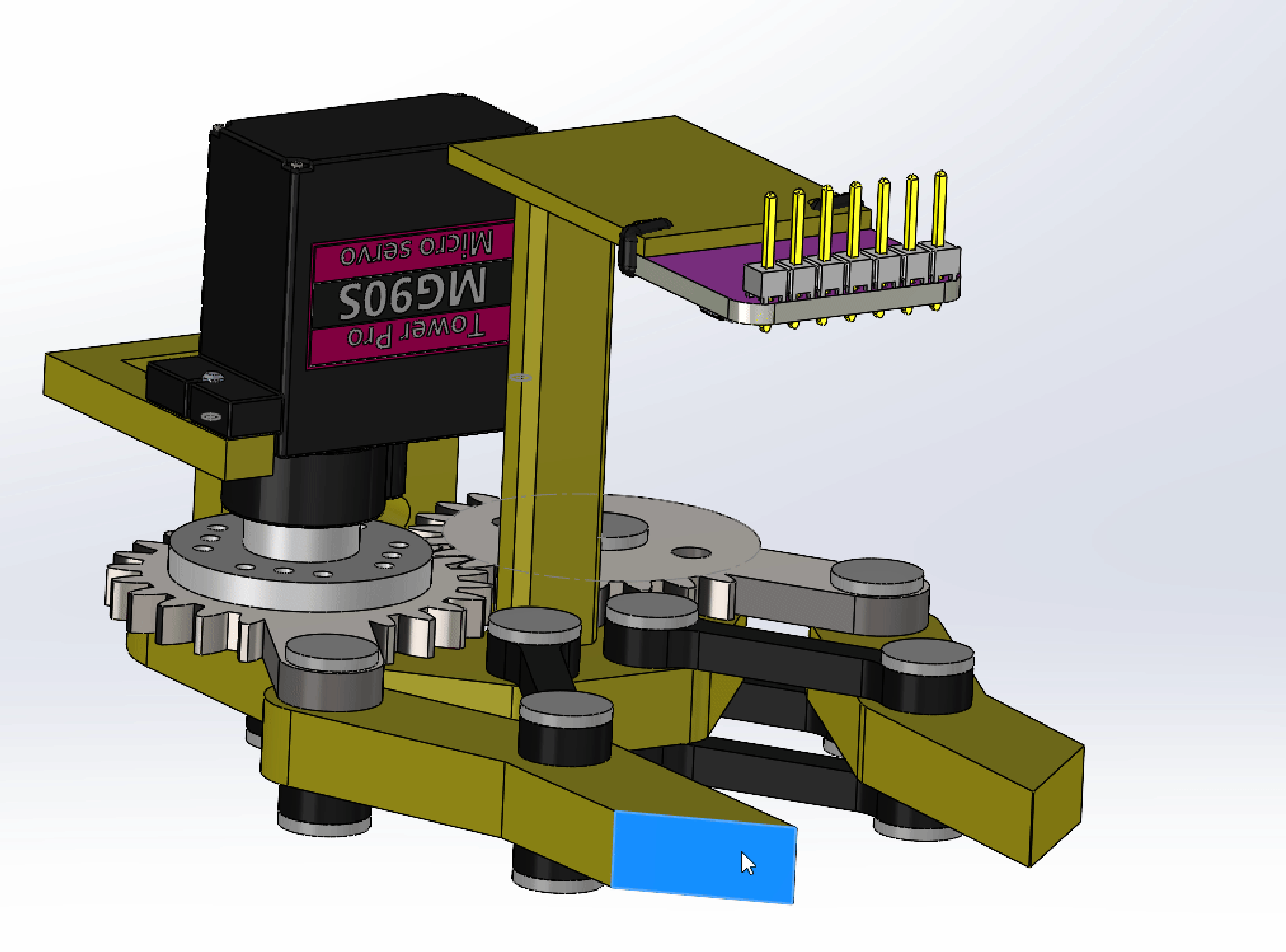
Rotary Axis 2: Deflection at End of Link

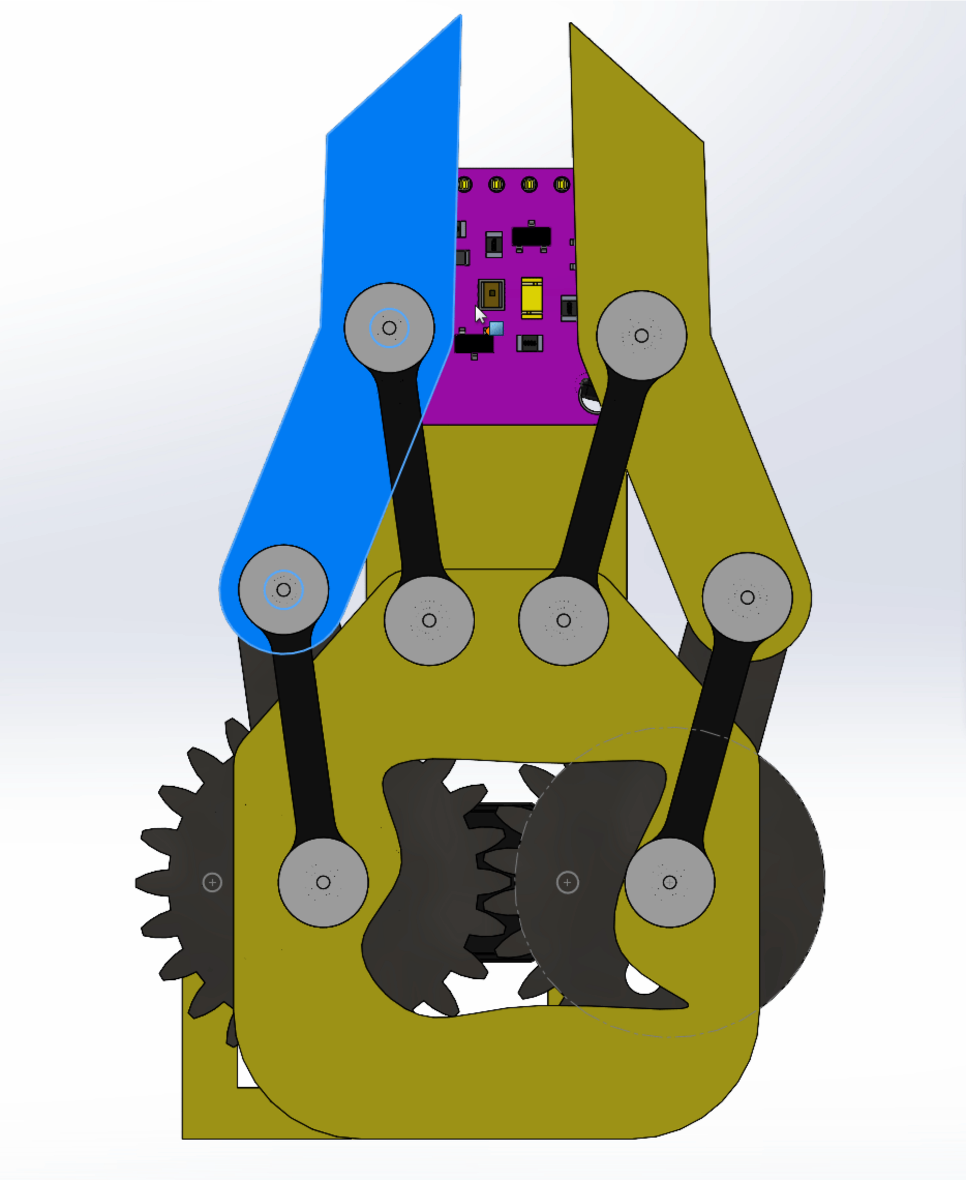
Component	Deflection Value
Link 1 (End of RA1)	0.621 mm
Link 2 (End of RA2)	0.0821 mm
Total Deflection at end of RA2	0.7031 mm

Rotary Axis 2: Torque and Resolution

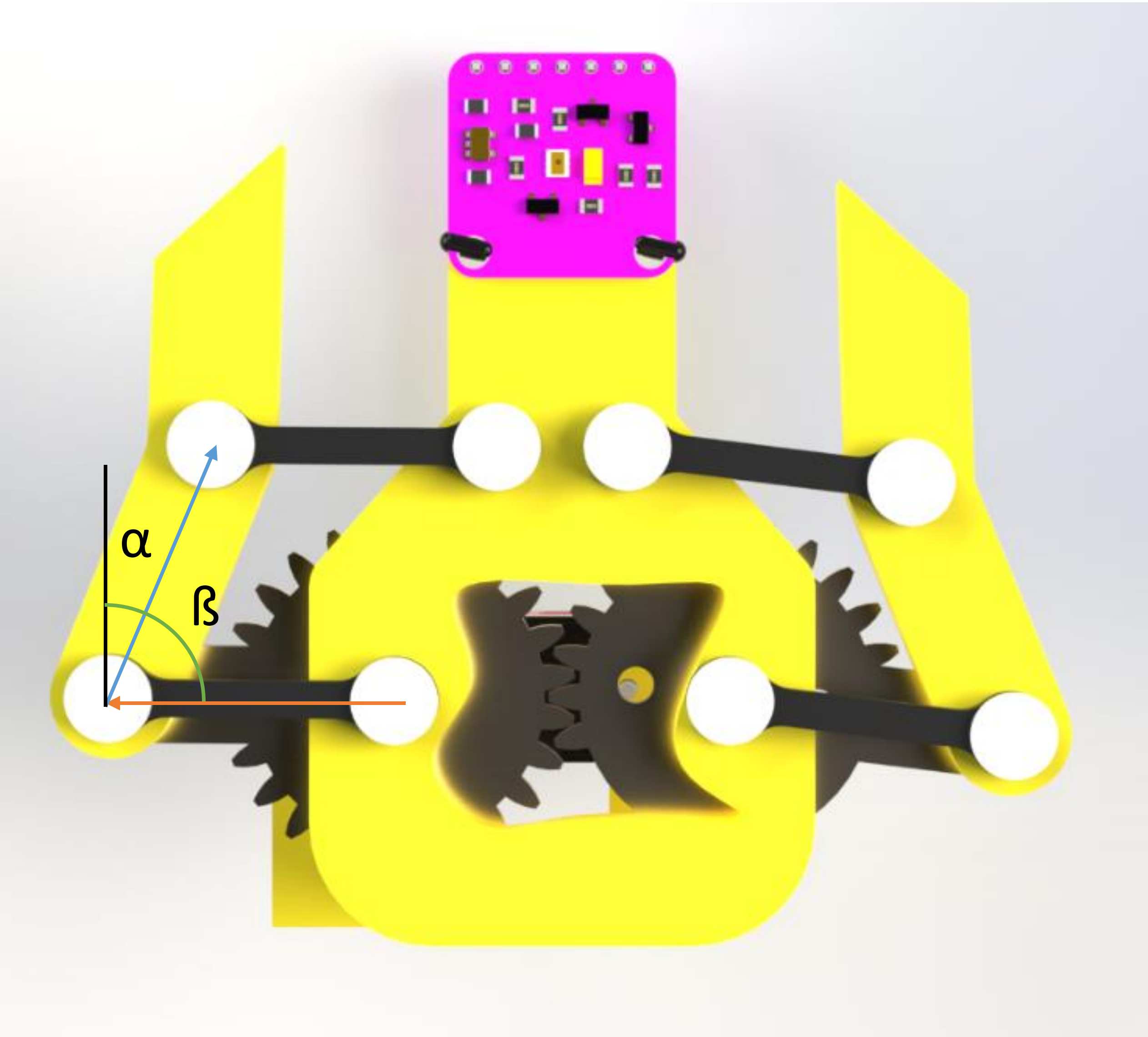
Resolution at End Effector	0.113 cm
Minimum Required Torque	24.16 Ncm
Torque Provided by Actuator	210.92 Ncm
Torque Safety Factor	8.73

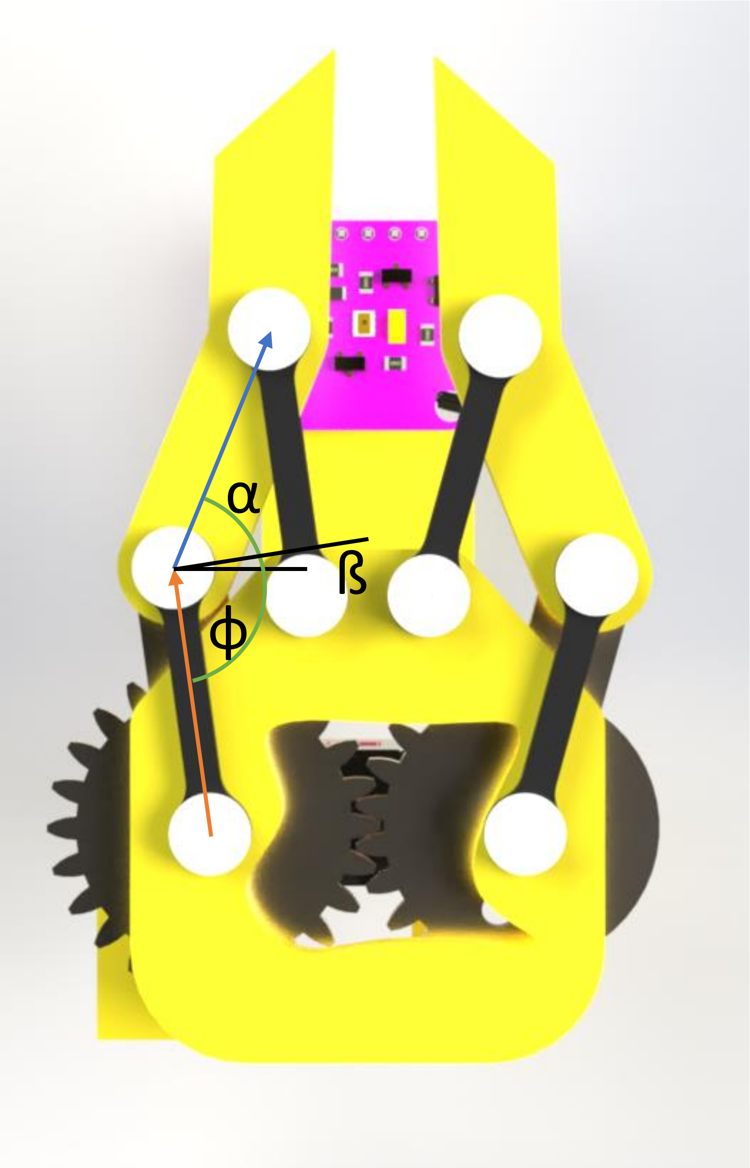
Gripper





Gripper: Open/Close Positions





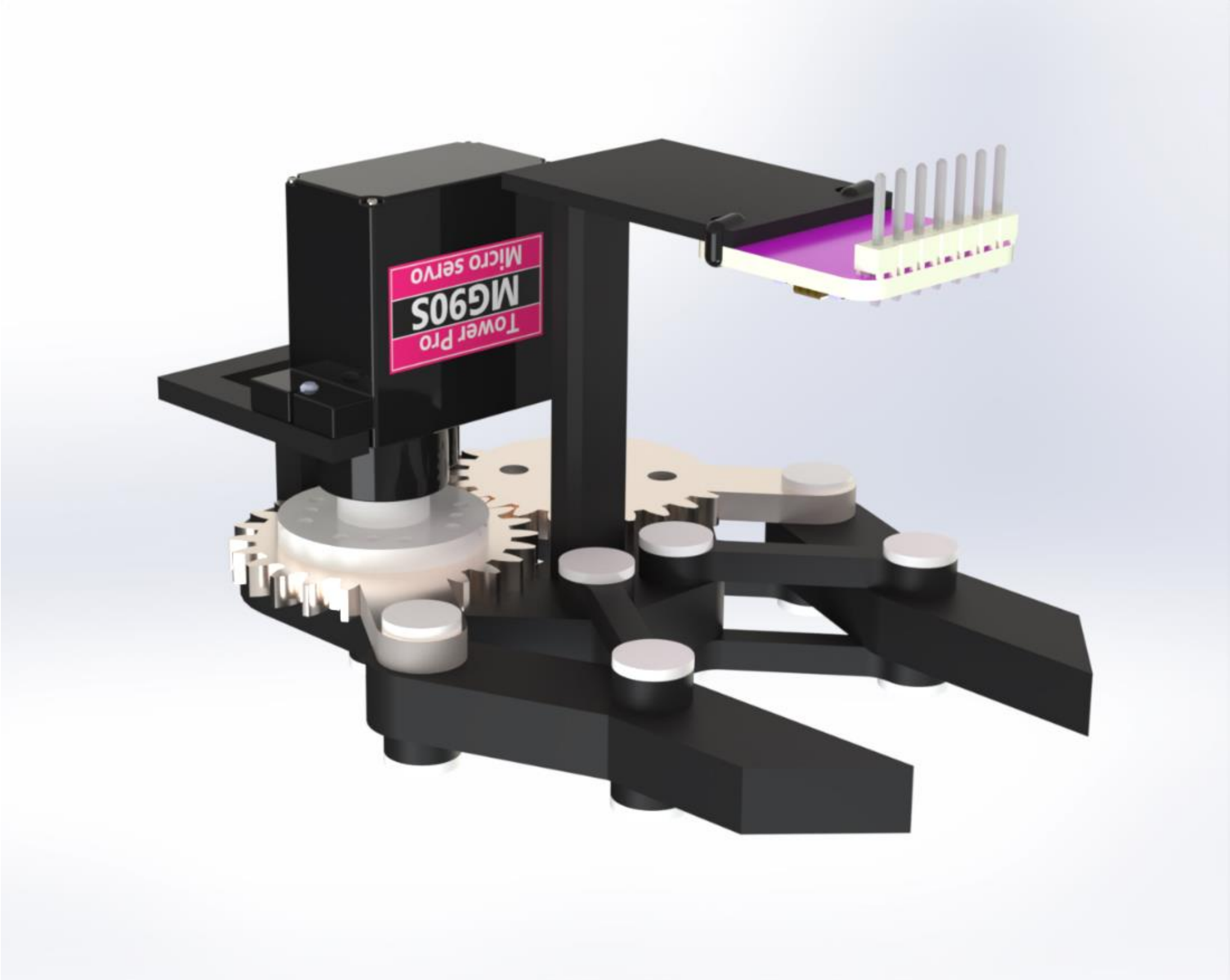
Torque required for open: 0.0569 Nm

Torque required for closed: 0.01507 Nm

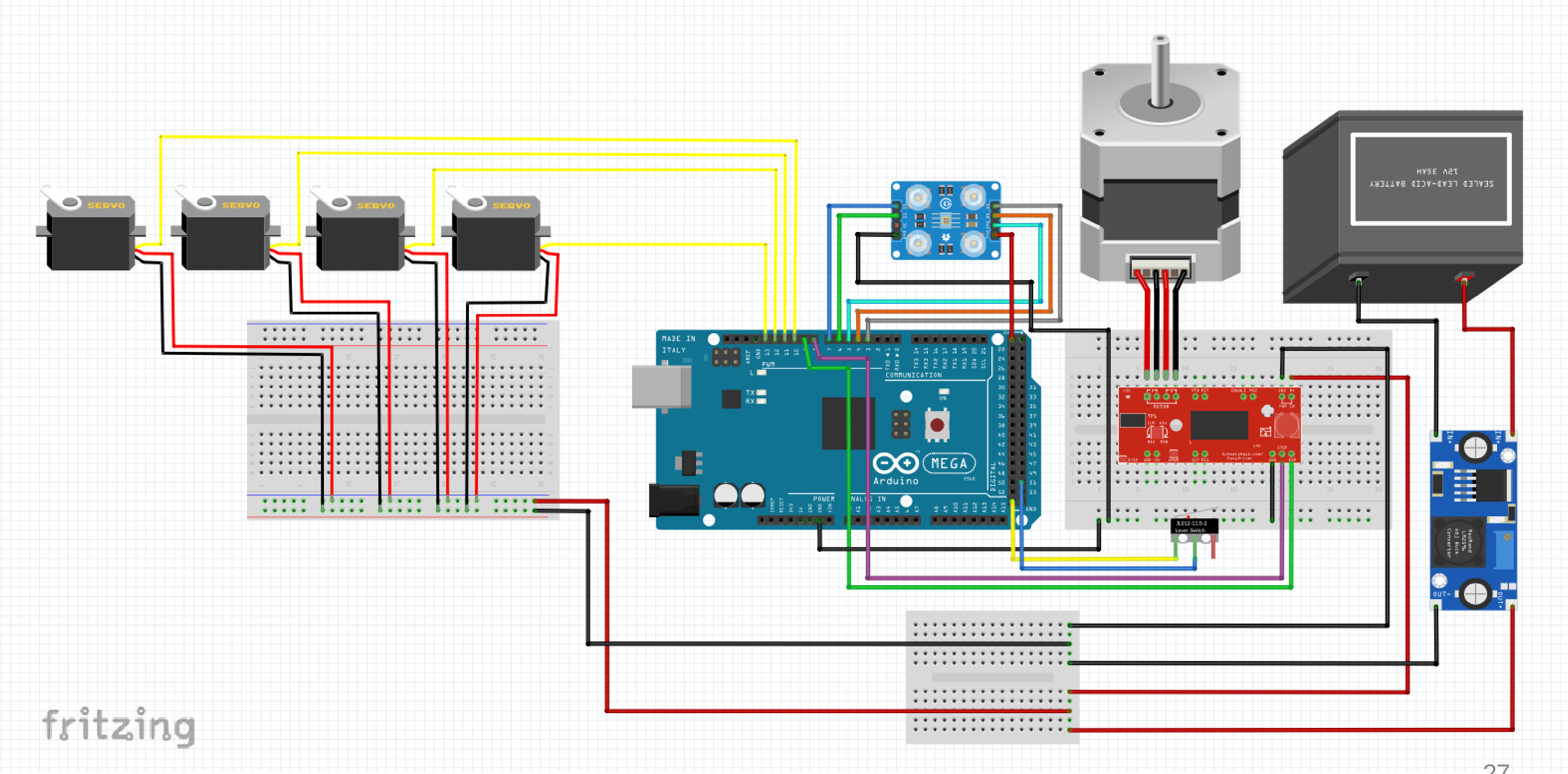
Mass: 106.2g

Range of Motion: 82.46°

Gripper: Final



Autonomous Controls: Wire Diagram



Autonomous Controls: Components List / Laptop Communication

Component	Quantity
Arduino Mega 2560 Rev3	1
TCS34275 Color Sensor	1
Nema 17 Stepper Motor	1
12V Power Supply	1
LM2596 Buck Converter	1
SparkFun EasyDriver Stepper Motor Driver	1
RDS3235 35kg Servo Motor	1
RDS3225 25kg Servo Motor	1
LDX-218 17kg Servo Motor	1
Connecting Wires	≈40

- Laptop will be connected to Arduino Mega 2560 Rev3, thus connected to everything via USB Type-B Cable

Autonomous Controls: Power Estimates

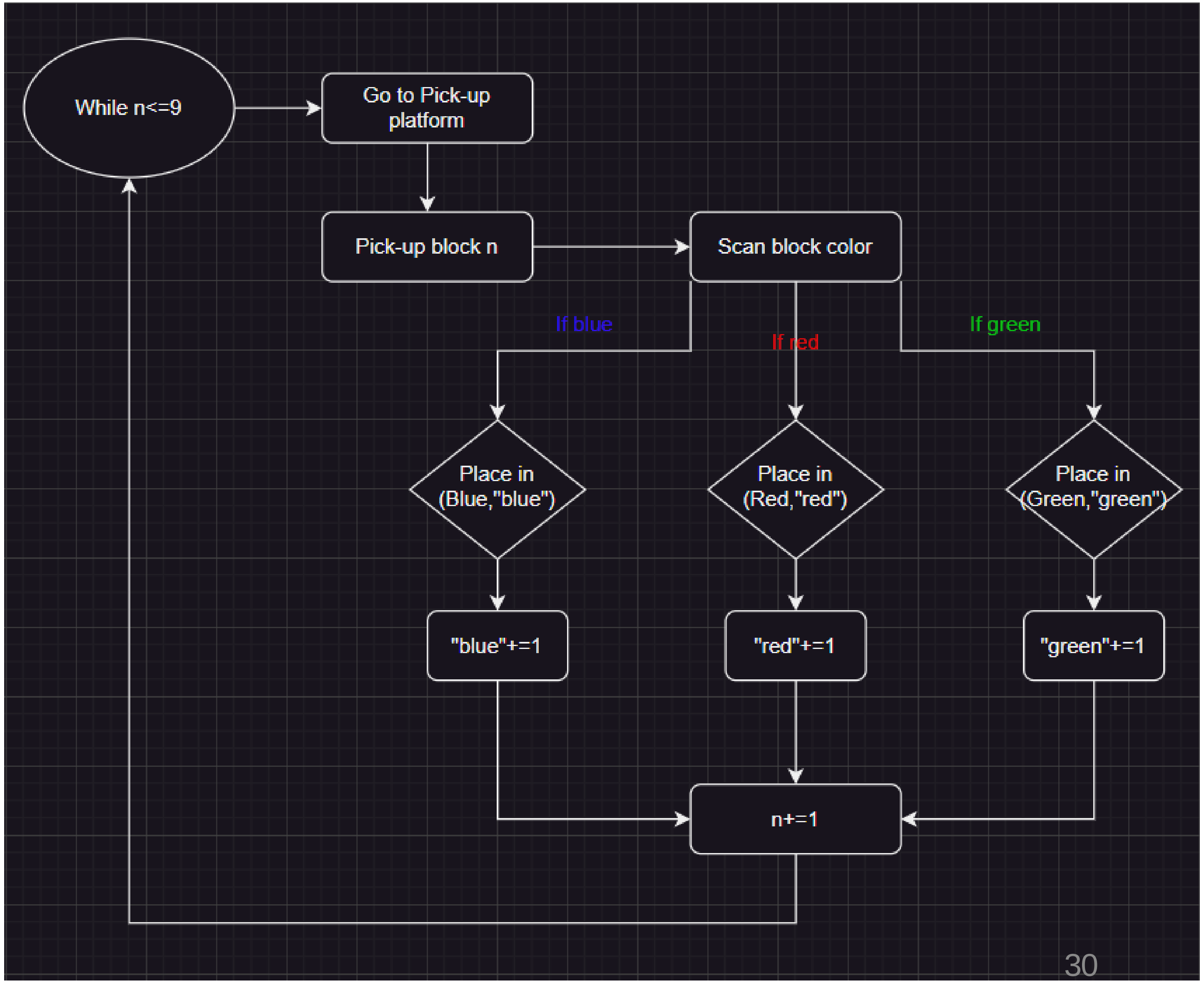
$$P = VI = \frac{V^2}{R} = I^2 R$$

Components	Final Power
Nema 17 Stepper Motor	18W
TCS34725 Color Sensor	1.65mW
RA1 Servo Motor	13.3W
RA2 Servo Motor	20.3W
Gripper Servo Motors	14.7W
Stepper Motor Driver	12.6W

	Totals
Amperage	11.9003A
Power	78.90165W

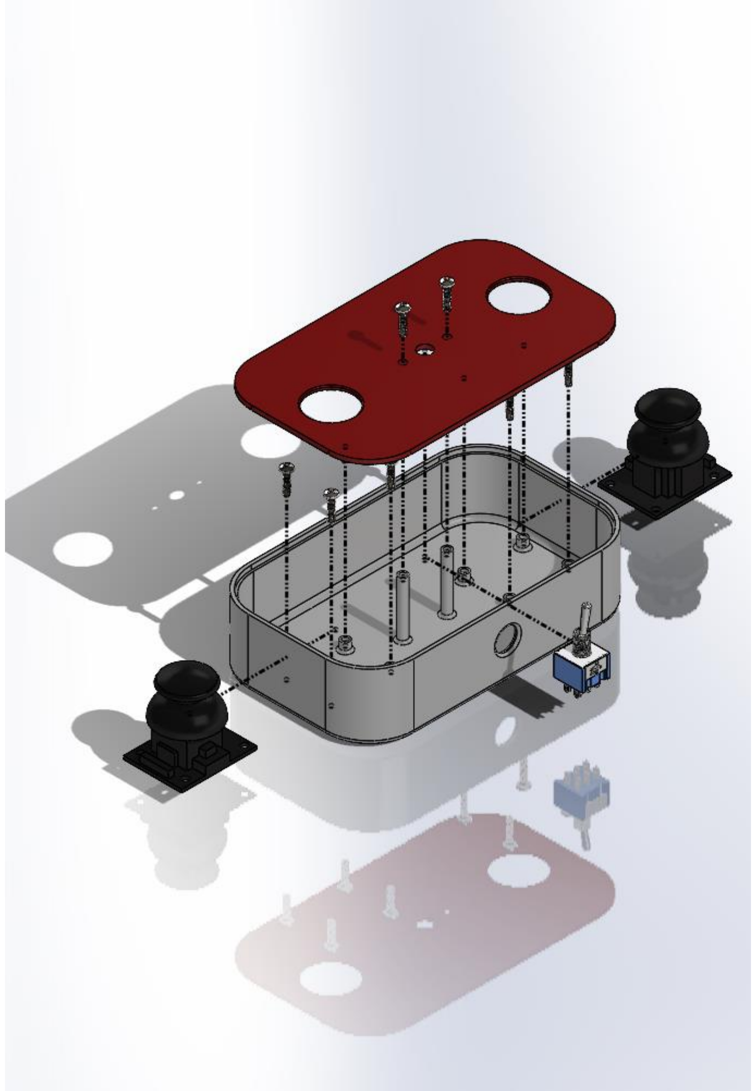
Autonomous Controls: Code Flowchart

- Pick-up location indexed 1-9
- X-value coordinates are the colors (Blue, Red, Green), y-value coordinates are the bins under that color
- Variable n is initialized to 1 and represents number of blocks
- Parameters “blue”, “red”, and “green” are initialized to 1 and represent current y-coordinate target
- Blocks are then scanned and placed into the correct spots using coordinates (Color, "color”)
- Once block is placed, n, “blue”, “red”, and “green” are incremented by 1

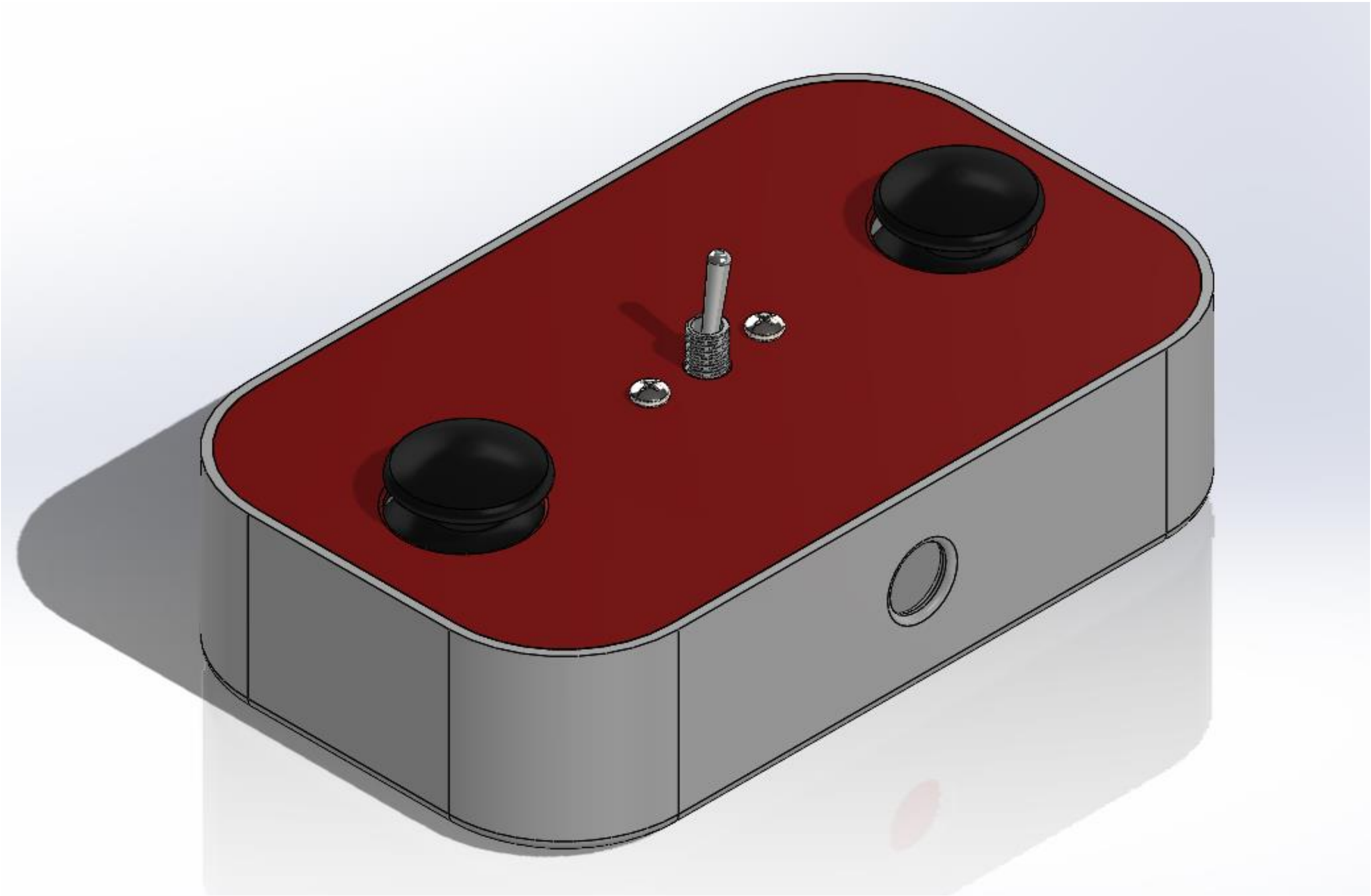


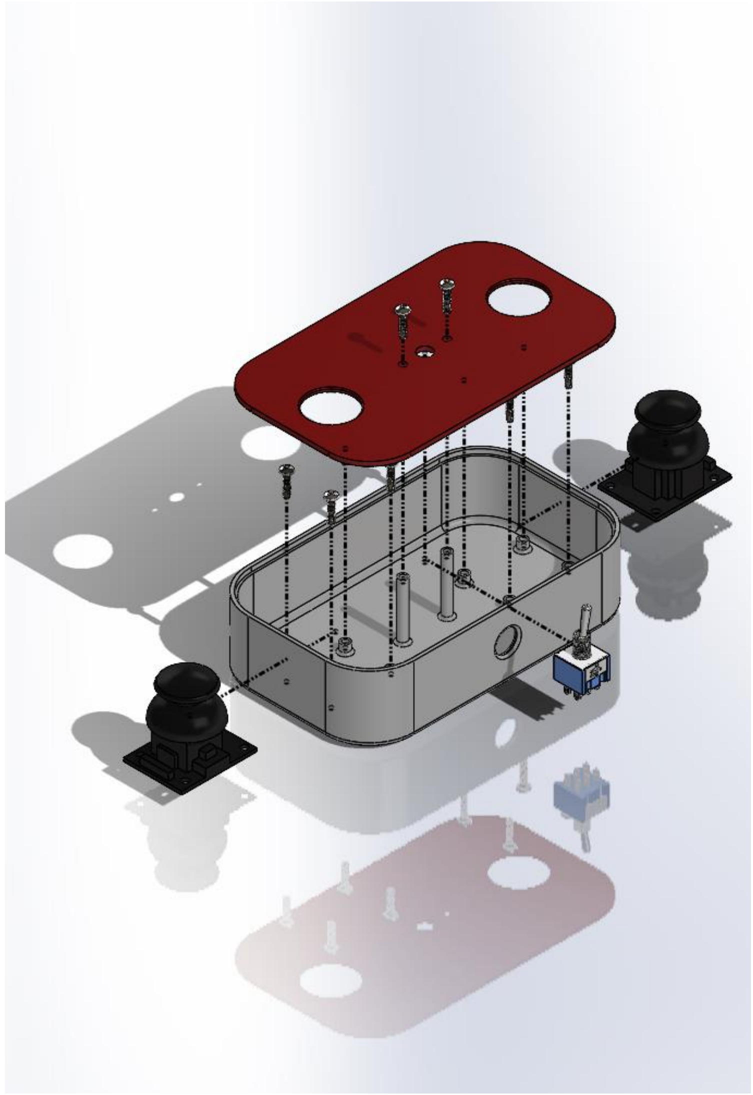
Manual Controls: Components

Component	Quantity
Analog 2-Axis Joystick	2
Toggle Switch	1
Controller Base	1
Controller Lid	1
Half Breadboard	1
Connecting Wires	≈ 16
M3 Screws	10
10 kΩ Resistor	2



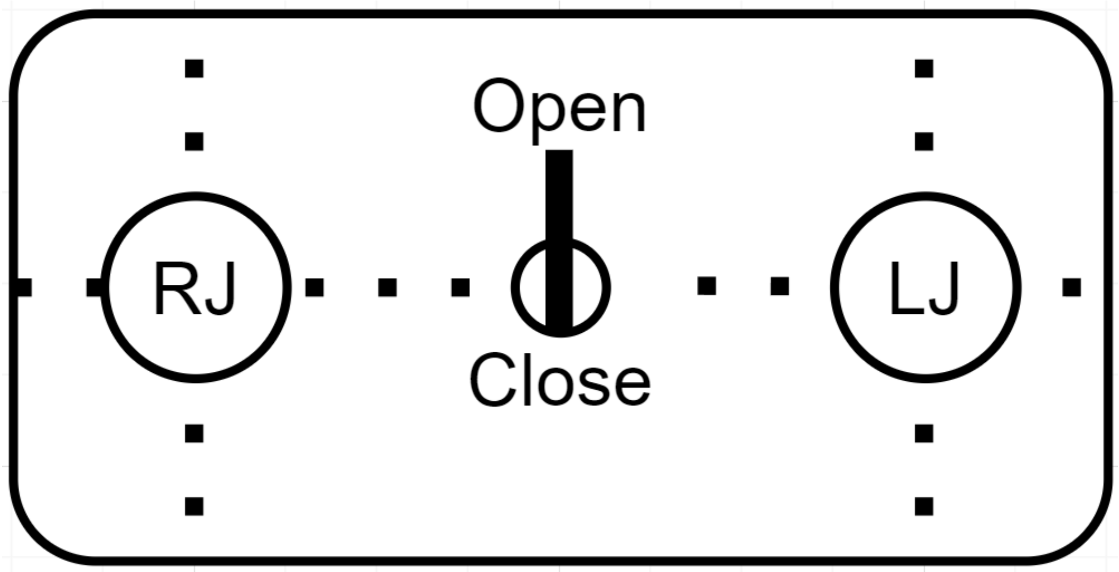
Manual Controls: Model



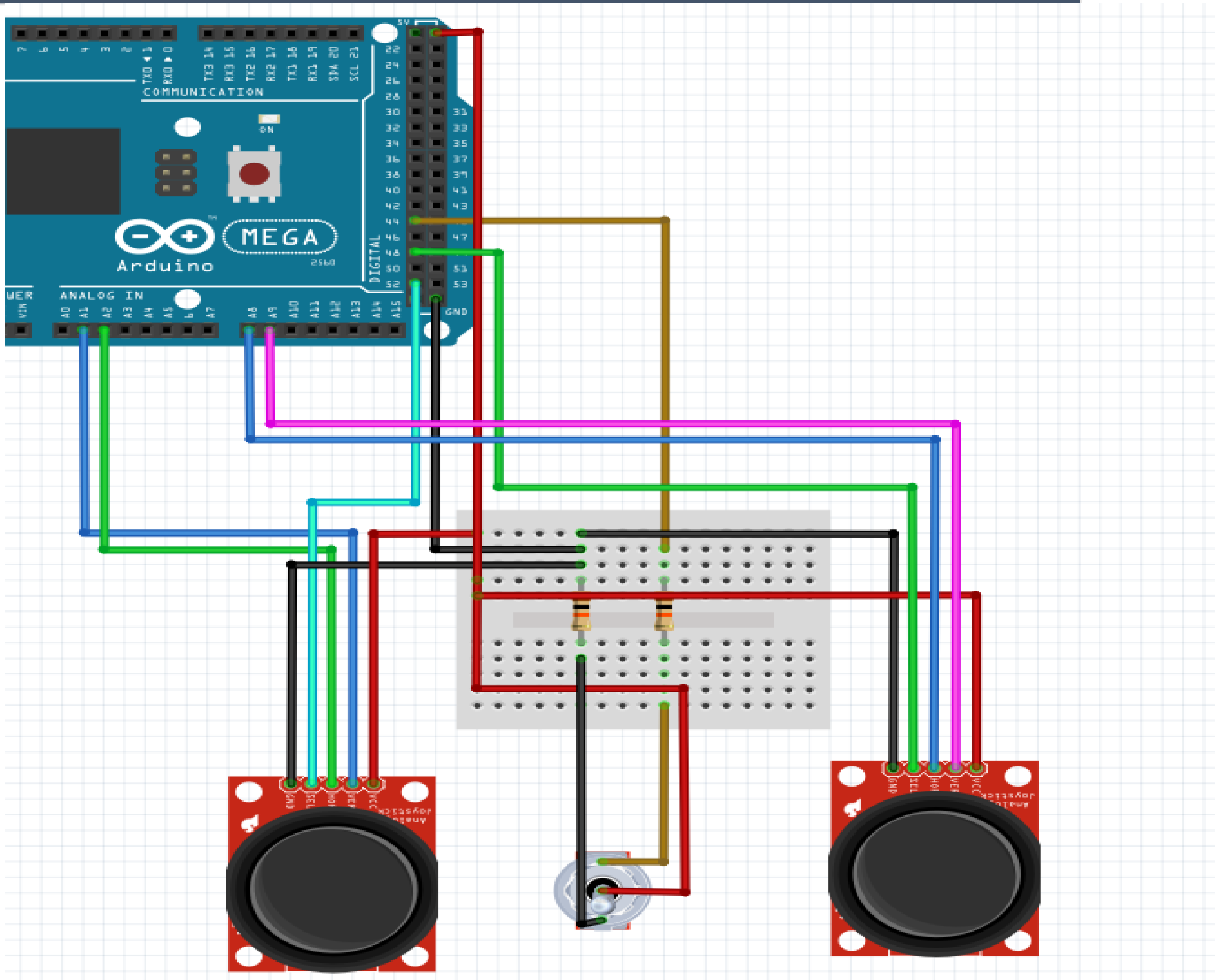


Manual Controls: Controls

Axis	Control
Linear Axis (Coarse)	RJ- X Axis
RA1	RJ- Y Axis
RA2	LJ- Y Axis
Linear Axis (Fine)	LJ- X Axis
Gripper	Open
Gripper	Close
Toggle Controls	Hold Both Joysticks (1s)



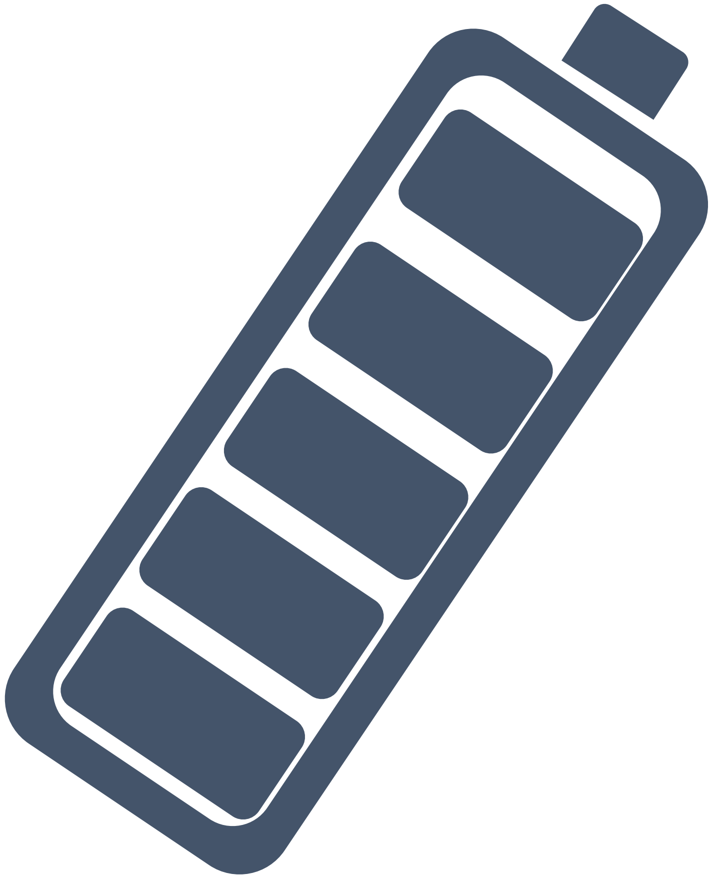
Manual Controls: Wiring



Manual Controls: Power

Component	Power
Toggle Switch	2.5mW
Joystick	10mW

	Total
Amperage	1.5mA
Power	12.5mW



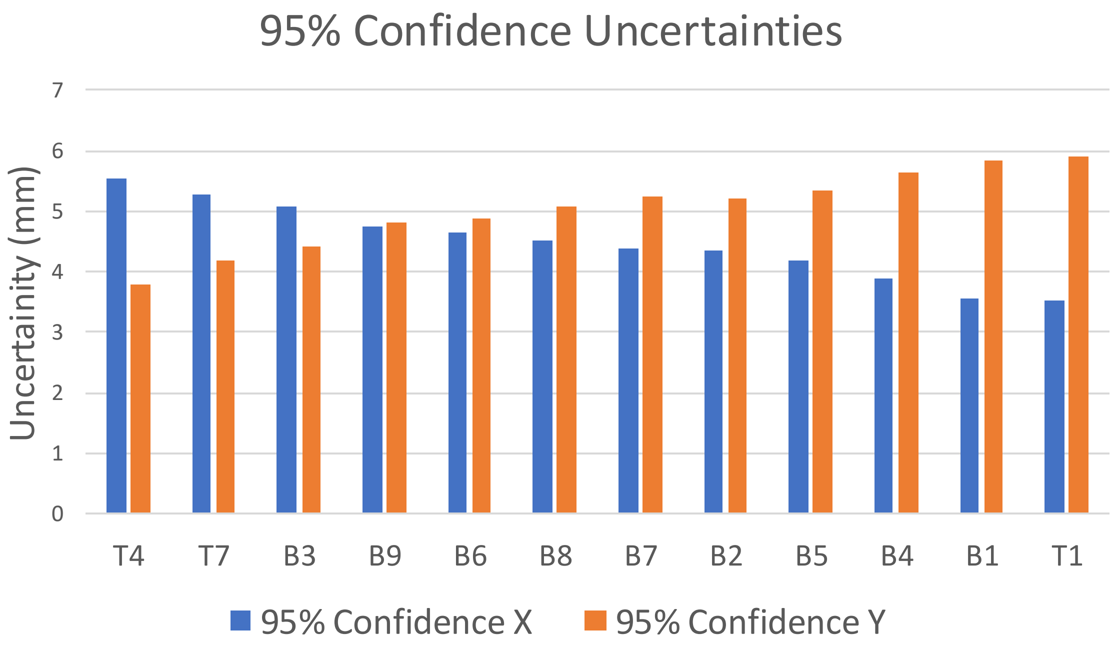
System: Mass

Subsystem	Mass
Base	168 g
Linear Axis	19,330 g
RA 1	101.5 g
RA 2	72.1 g
Gripper	106.2 g
Total	19,777.8 g

System: Positioning Uncertainty

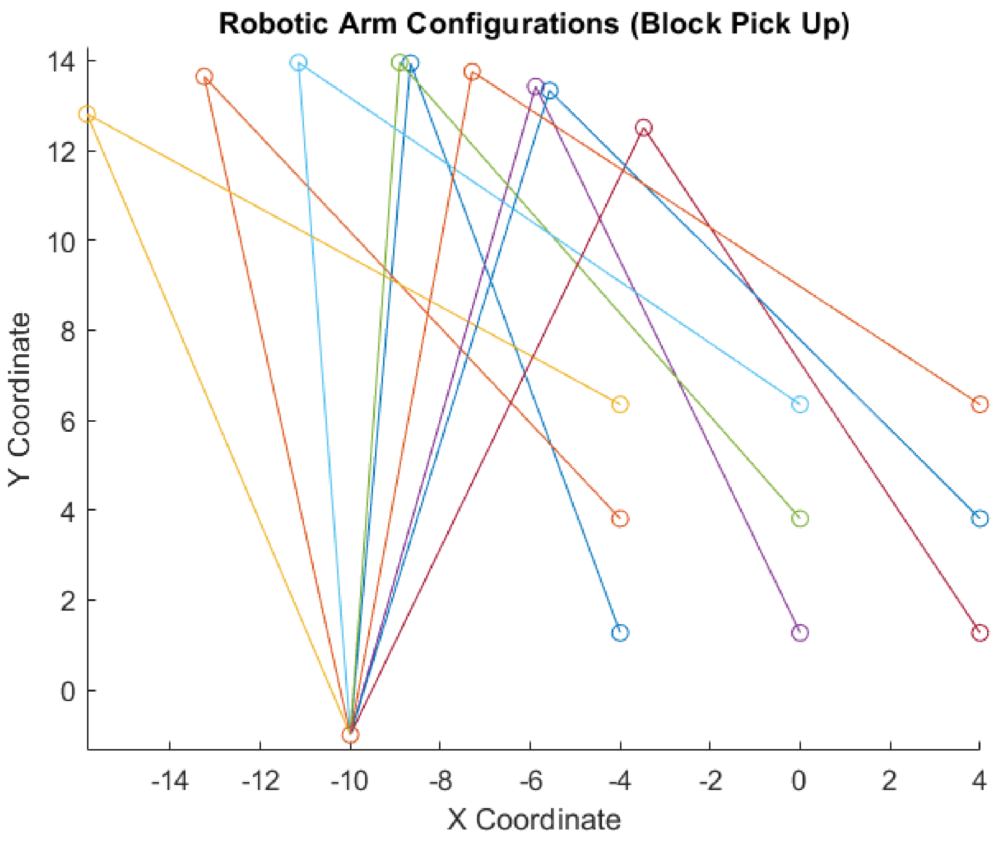
- Kinematic Equations
- $A(x) = L_1 cos(\theta_1) + L_2 cos(\theta_1 + \theta_2) + L_3 cos(\theta_1 + \theta_2 + \theta_3)$
- $A(y) = L_1 sin(\theta_1) + L_2 sin(\theta_1 + \theta_2) + L_3 sin(\theta_1 + \theta_2 + \theta_3)$

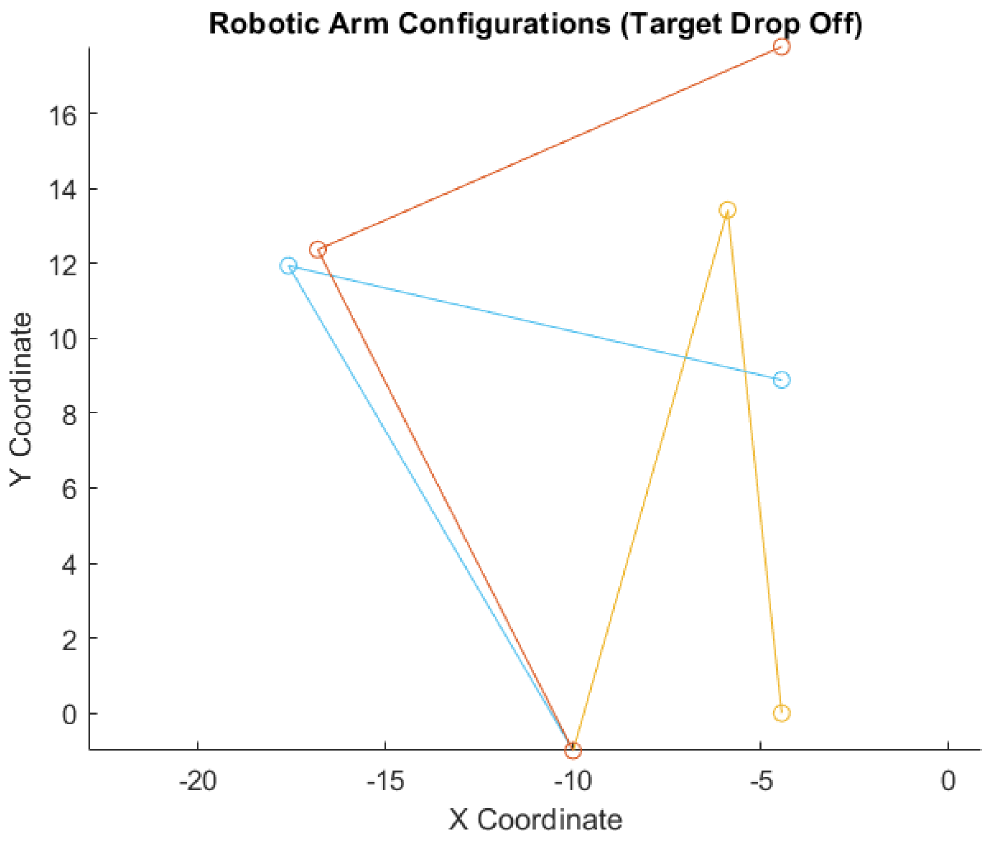
Uncertainty Inputs
Link Length = 0.56 mm
ϴ = 0.5°



Max Uncertainty X	5.54 mm (T4)
Max Uncertainty Y	5.92 mm (T1)
Average Uncertainty X	4.47 mm
Average Uncertainty Y	5.03 mm

System: Kinematic Results





System: Axis Homing

1. Microswitch will be placed at the x-z coordinate (-10,0)
2. Homing sequence will be set at the beginning of the autonomous code so that the coordinate system can be created for that specific sorting operation.
3. Every time the robot comes back to pick-up location, homing will be reset so that any uncertainty does not accumulate each time the robot performs a certainty operation

System: Autonomous Time Estimates

Motion	Time Estimate
Block Pick-Up & Place	45 seconds
Linear Axis Motion (Left-Most Column) – 3 Blocks	10 seconds
Linear Axis Motion (Middle Column) – 3 Blocks	6.672 seconds
Linear Axis Motion (Right-Most Column) – 3 Blocks	3.336 seconds
Total Time	**65.003 seconds**

- 9 block pick-ups & placements = 45 seconds
- 3 blocks moved to the left-most column = 3 * 3.333 seconds = 10 seconds
- 3 blocks moved to the middle column = 3 * 2.224 seconds = 6.672 seconds
- 3 blocks moved to the right-most column = 3 * 1.112 seconds = 3.336 seconds
- Total time will be ≈ 1 minute & 5 seconds

Use of Rapid Prototyping

- The Following Components will be Rapid Prototyped:
    1. Link 1
    2. Link 2
    3. Gripper Structure (Body, Effectors, Punches, Gears)
    4. Manual Controller Container (Base, Top)

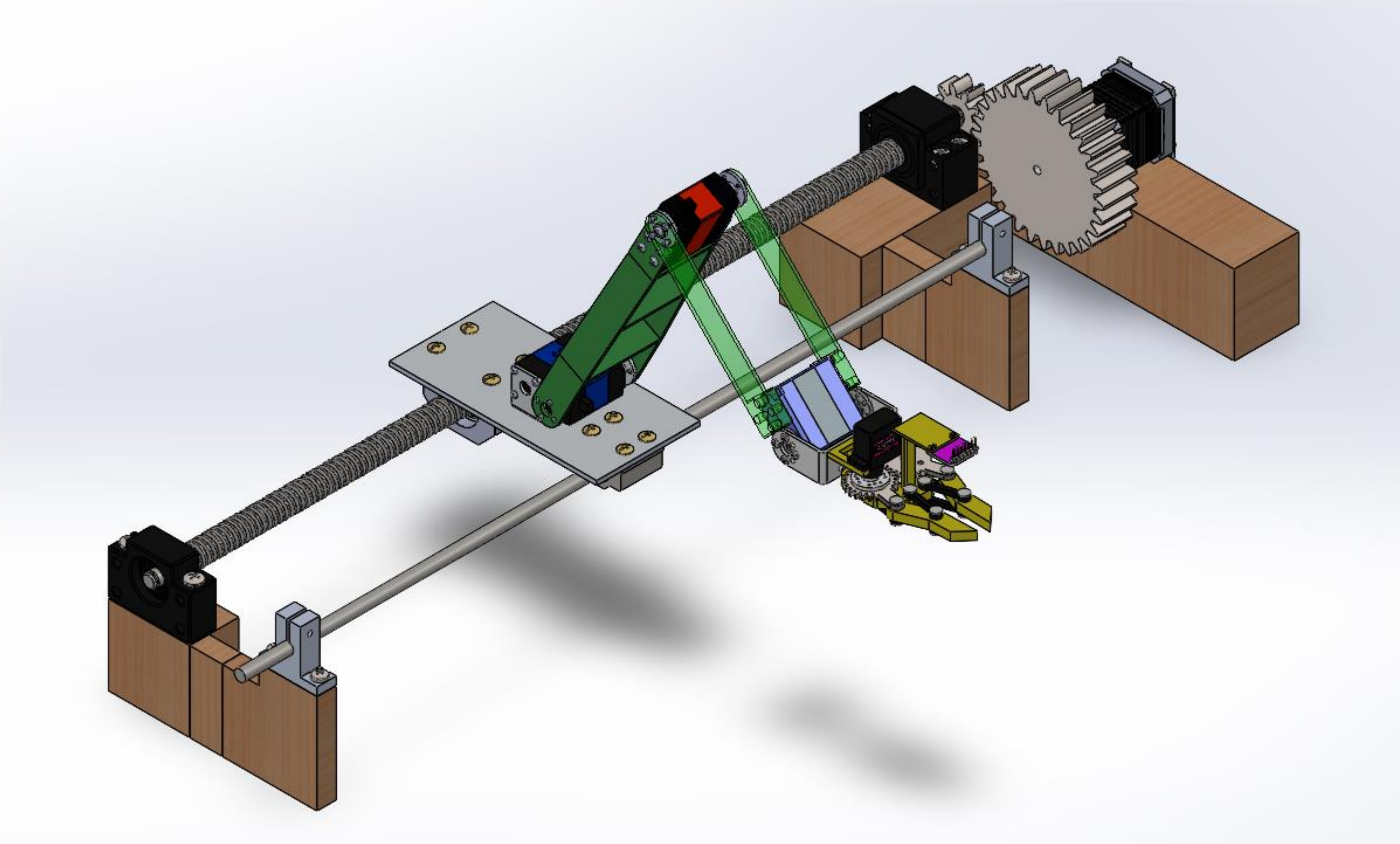
Mechanism Manufacturability

Subsystem	Component	Manufacturability
Base	Custom machined base	Manufactured (Machine Shop)
	M3 and M4 Screws	Purchased
Linear Axis	NEMA 17 Stepper	Purchased
	Linear Motion Shaft Kit	Purchased
	Base Supports (Wood)	Manufactured (Maker Space)
	Lead Screw	Purchased
	Microswitch	Purchased
RA 1	Servo Motor Assembly	Purchased
	Link 1	Manufactured (3D Print)
RA 2	Servo Motor Assembly	Purchased
	Link 2	Manufactured (3D Print)
Gripper	Servo Motor Assembly	Purchased
	Color Sensor	Purchased
	Servo (Gripping)	Purchased
	Body/Punches/Effectors/Gears	Manufactured (3D Print)

Budget

Subsystem	Component	Material/Part Cost	Qty	Labor Cost	Total	Subsystem Total
Base	Custom Machined Base	$ 26.11	1	$ 20.00	$ 46.11	
	M3 Screws	$ 0.02	4		$ 0.08	
	M4 Screws	$ 0.02	4		$ 0.08	
	M5 Screws	$ 0.02	4		$ 0.08	$ 46.35
Linear Axis	Nema 17	$ 8.99	1		$ 8.99	
	Linear Motion Shaft Kit	$ 31.99	1		$ 31.99	
	Base Support	$ 10.55	1		$ 10.55	
	Lead Screw Assembly	$ 46.88	1		$ 46.88	
	Microswitch	$ 2.95	1		$ 2.95	$ 101.36
RA 1	Servo Motor Assembly	$ 36.59	1		$ 36.59	
	3D Printed Link	$ 6.84	1		$ 6.84	$ 43.43
RA 2	Servo Motor Assembly	$ 21.99	1		$ 21.99	
	3D Printed Link	$ 1.03	1		$ 1.03	$ 23.02
Gripper	Actuation Servo	$ 18.99	1		$ 18.99	
	Color Sensor	$ 6.99	1		$ 6.99	
	Gripper Servo	$ 7.39	1		$ 7.39	
	Heat Set Inserts	$ 0.06	4		$ 0.24	
	3D Printed Gripper	$ 1.20	1		$ 1.20	$ 34.81
Controls	Arduino	$ 51.91	1		$ 51.91	
	Pack of Wires	$ 12.00	1		$ 12.00	
	Toggle Switch	$ 8.00	1		$ 8.00	
	3D Printed Controller	$ 15.70	1		$ 15.70	
	Joysticks [2 pack]	$ 13.00	1		$ 13.00	
	Buck Converter	$ 5.49	1		$ 5.49	
	Stepper Motor Driver	$ 18.17	1		$ 18.17	
	Zip Ties	$ 3.00	1		$ 3.00	$ 127.27
Materials Subtotal						$ 376.24
Student Hours	Design Hours		180	$ 10.00	$ 1,800.00	
	Assembly Hours		120	$ 10.00	$ 1,200.00	
	Testing Hours		75	$ 10.00	$ 750.00	
Labor Subtotal						$ 3,750.00
Robot Total						$ 4,126.24

Summary



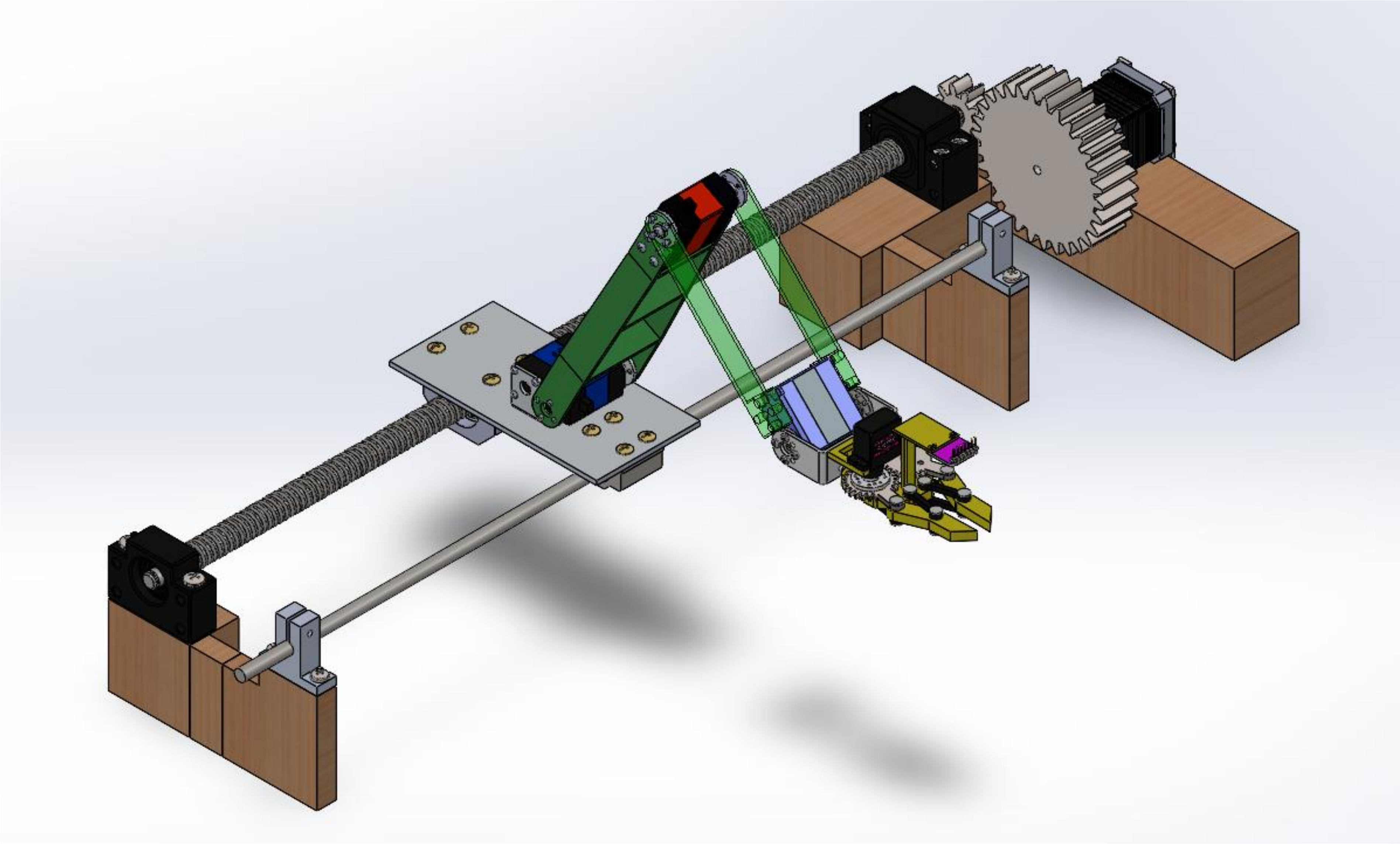
References

- 35 kg servo motor CAD File: https://grabcad.com/library/rds-3235-servo-1
- Cantilever Beam Equations: Mechanics of Materials (5th edition) Philpot and Thomas
- RDS 3225 CAD File: https://grabcad.com/library/rds-3225-servo-1
- MG996R CAD File: https://grabcad.com/library/mg996r-servo-motor-3
- Analog 2-axis joystick https://www.autodesk.com/community/gallery/project/156442/arduino-joystick
- Toggle switch https://www.autodesk.com/community/gallery/project/154819/toggle-switch---onon---2-pole---6a
- Micro Switch https://forum.fritzing.org/t/jl012-13-5-2-mini-lever-limit-switch/18373
- Color Sensor https://forum.fritzing.org/t/color-sensor-tcs230-color-sensor/14922
- Stepper Motor Driver https://www.sparkfun.com/products/retired/11876#:~:text=Each%20Big%20Easy%20Driver%20can,only%20one%20supply%20is%20necessary.

References

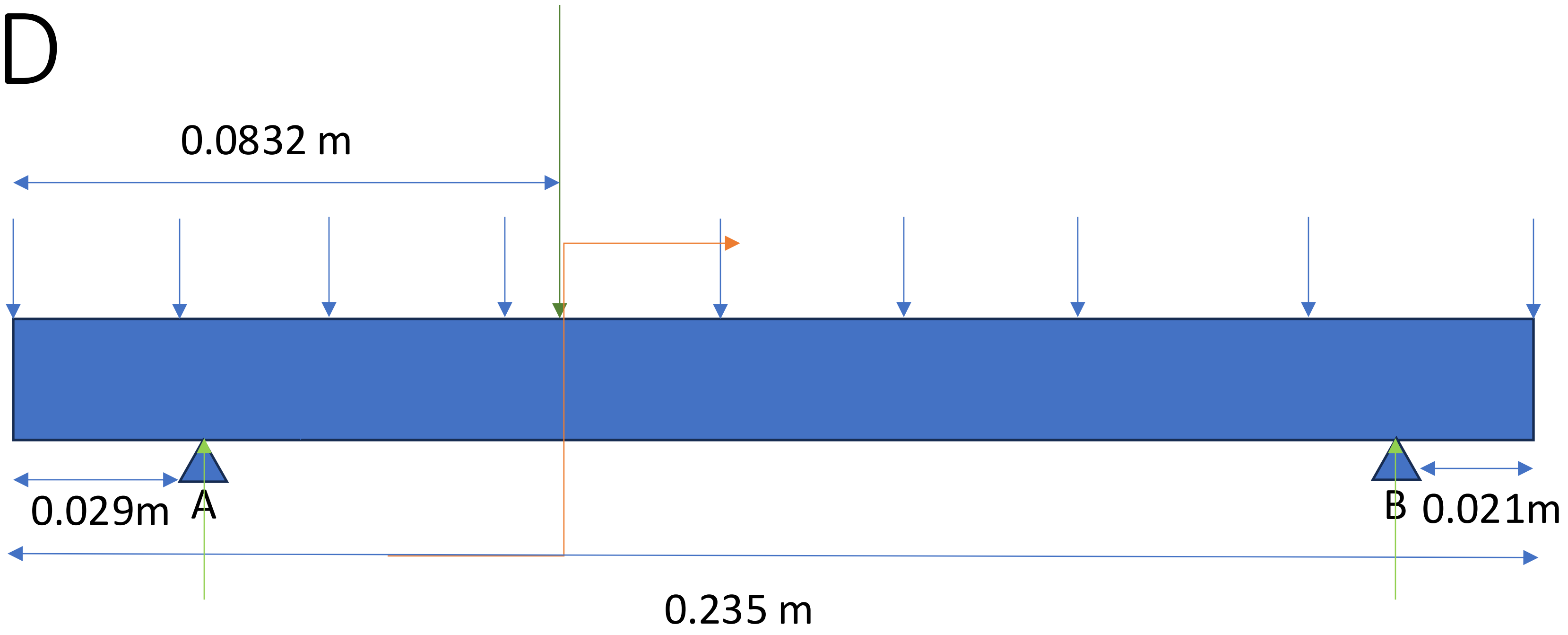
•Stepper Motor https://catalog.orientalmotor.com/viewitems/2-phase-bipolar-stepper-motors/42mm-pkp-series-2-phase-bipolar-stepper-motors?gclid=CjwKCAiA_OetBhAtEiwAPTeQZ0tdlJgoD-rX9CKwAHlTpvDbmU4BuFAEJqc7UAVqbnyqo9mxXoRByxoC6coQAvD_BwE
•Color Sensor Power Rating https://cdn-shop.adafruit.com/datasheets/TCS34725.pdf
•Easy Driver - Arduino Connection https://forum.arduino.cc/t/problem-using-arduino-mega-as-cnc-driver/390716
•Servo Motor - Arduino Connection https://wiki-content.arduino.cc/en/Tutorial/LibraryExamples/Sweep
•Color Sensor - Arudino Connection https://lastminuteengineers.com/tcs230-tcs3200-color-sensor-arduino-tutorial/

Questions?



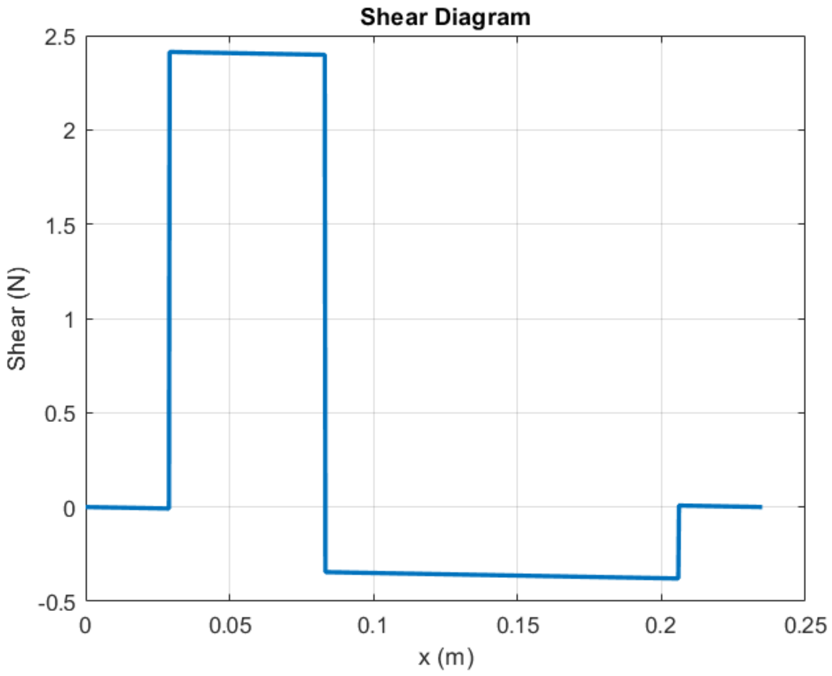
Appendix

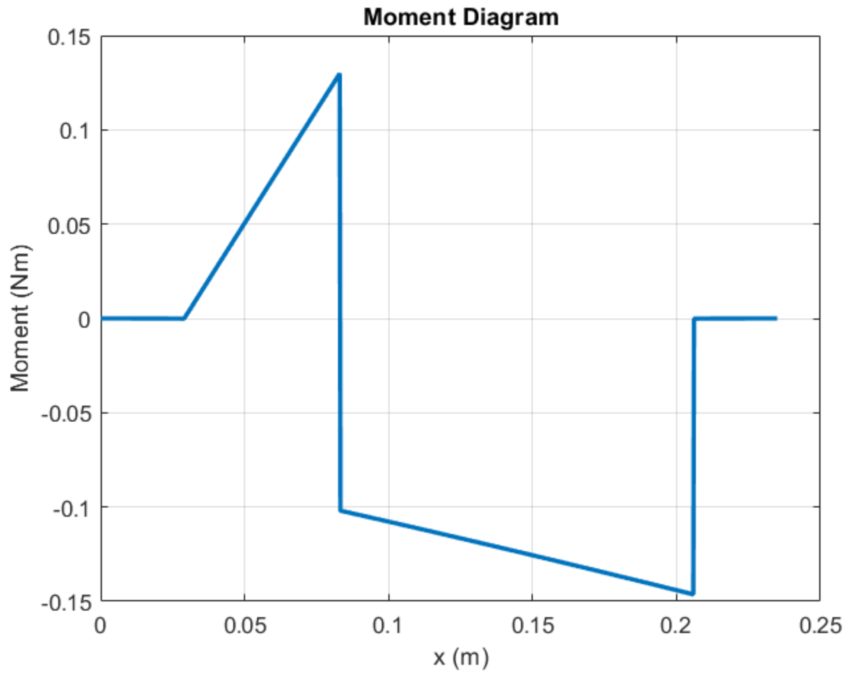
Base FBD



Marker	Value
Distributed load (blue)	4.08 N/m
Point load (green)	2.744 N
Moment (orange)	-43.05 Nm
Reaction at A	2.44 N
Reaction at B	1.26 N

Base Shear-Moment Diagrams





Distance (m)	Equation
0 – 0.0290	-0.277x
0.0290 – 0.0832	-0.277x + 2.421
0.0832 - 0.206	-0.277x – 0.322
0.2060 – 0.2350	-0.277x + 0.065

Distance (m)	Equation
0 – 0.0290	$-0.113x^2$
0.0290 – 0.0832	$-0.113x^2 + 2.421x - 0.070$
0.0832 - 0.206	$-0.113x^2 - 0.322x - 0.074$
0.2060 – 0.2350	$-0.113x^2 + 0.065x - 0.008$

Base Bending Stress

Variable	Description	Value
M	Max moment	-0.1487 Ncm
c	Distance from COG	0.15 cm
I	Moment of inertia	6.2678 x 10^-10

$$\sigma = \frac{Mc}{I}$$

```
bendstress =

    0.3766
```

(units in MPa)

Base Shear Stress

Variable	Description	Value
V	Max Shear stress	2.43 N
Q	Area times distance from centroid	0.0007 m^2
I	Moment of inertia	6.2678 x 10^-10 m^4
b	Width from neutral axis	0.0325 m

$$\tau = \frac{VQ}{Ib}$$

```
shear =

    0.0975
```

units in MPa

Base SF

Variable	Value
Von Mises Stress	0.3766 MPa
Yield strength (Al 6061)	276 MPa

$$SF = \frac{Yield\ Strength}{Von\ Mises\ Stress}$$

```
sf =

   732.8282
```

Base Deflection

Variable	Description	Value
W	Distributed load	4.08 N/m
P	Point load	2.744 N
d	right end of beam to right pin	0.021 m
E	Modulus of elasticity (6061 Al)	69 GPa
l	Distance between pins	0.185 m
L	Total length of beam	0.235 m
c	Distance from left end of beam to left pin	0.029 m
a	Distance from left pin to point load	0.054 m
b	Distance from right pin to point load	0.062 m

```
deflect =

   1.2168e-18
```

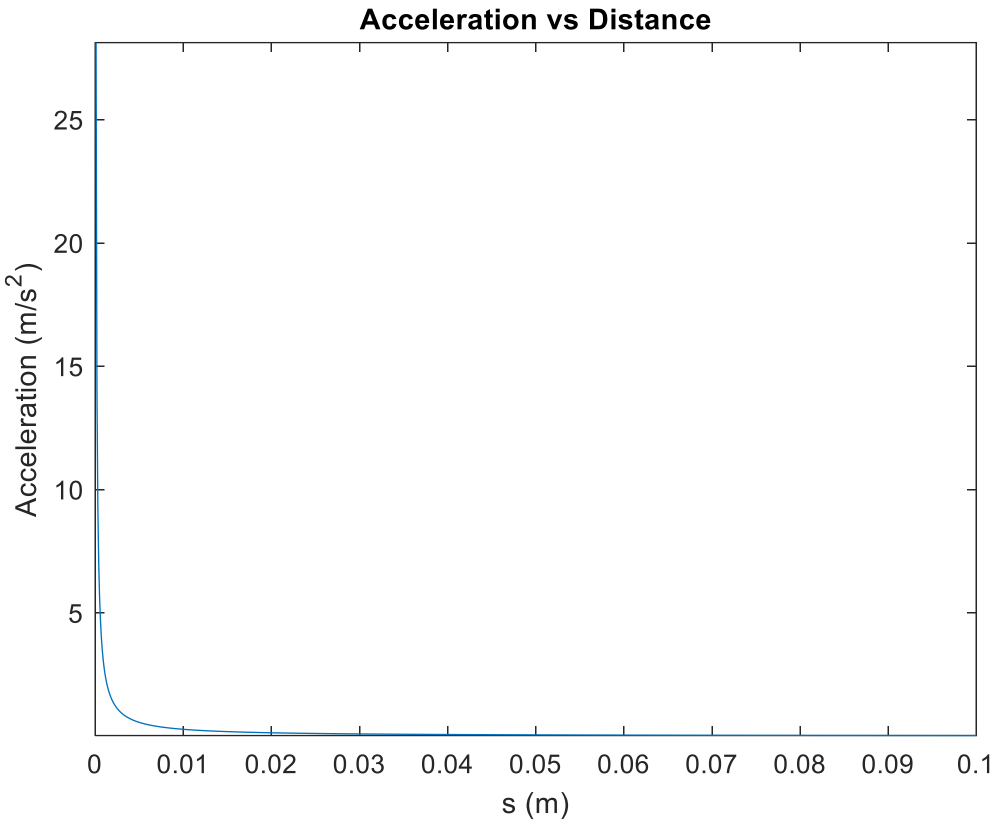
(units in m)

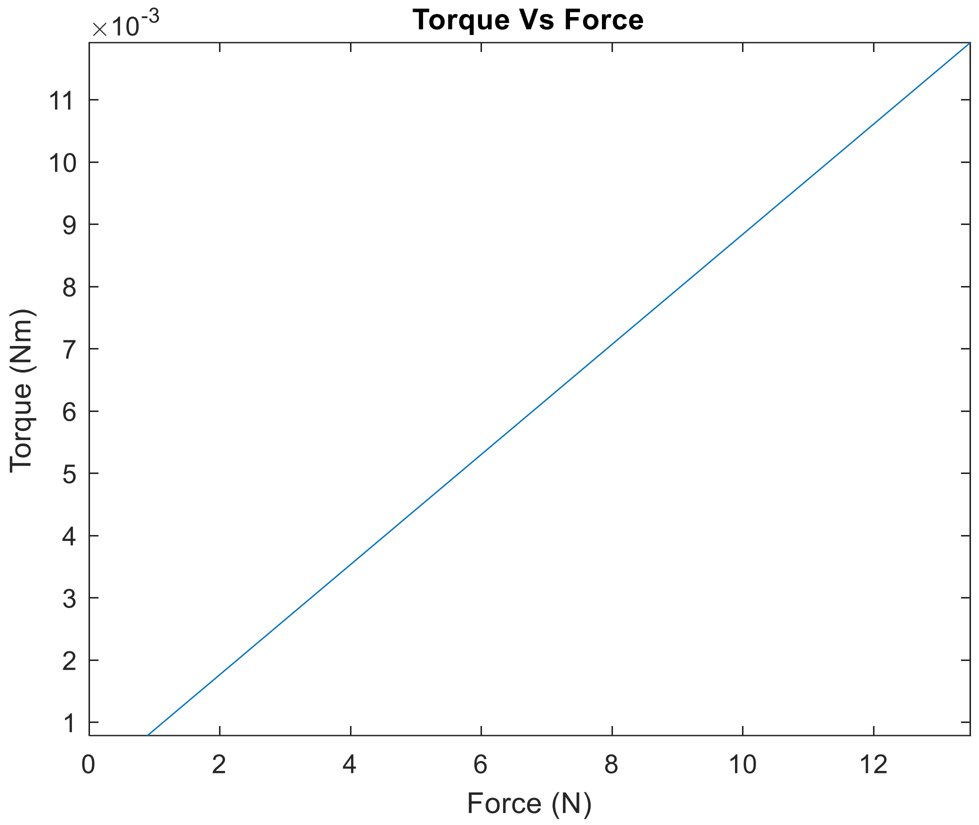
$$\delta = \frac{Wx(l-x)}{24EIL} * \left(x(l-x) + l^2 - 2(d^2 + c^2) - \left(\frac{2}{l}\right) * \left(d^2x + c^2(1-x)\right) + \frac{Wbx}{6EIl} * (d^2 - x^2 - b^2)\right.$$

Linear Axis Torque and Speed Calculations

Variable	Equation	Required Spec
T	$\frac{F * L}{2\pi e}(GR)$	.036Nm
F	$m(a + 2\mu g)$	13.5N
a	$\frac{v^2}{2s}$	28 m/s^2
L	Lead Screw Spec	5mm/rev
μ	bearing friction	0.1
m	All components above linear axis	.448kg

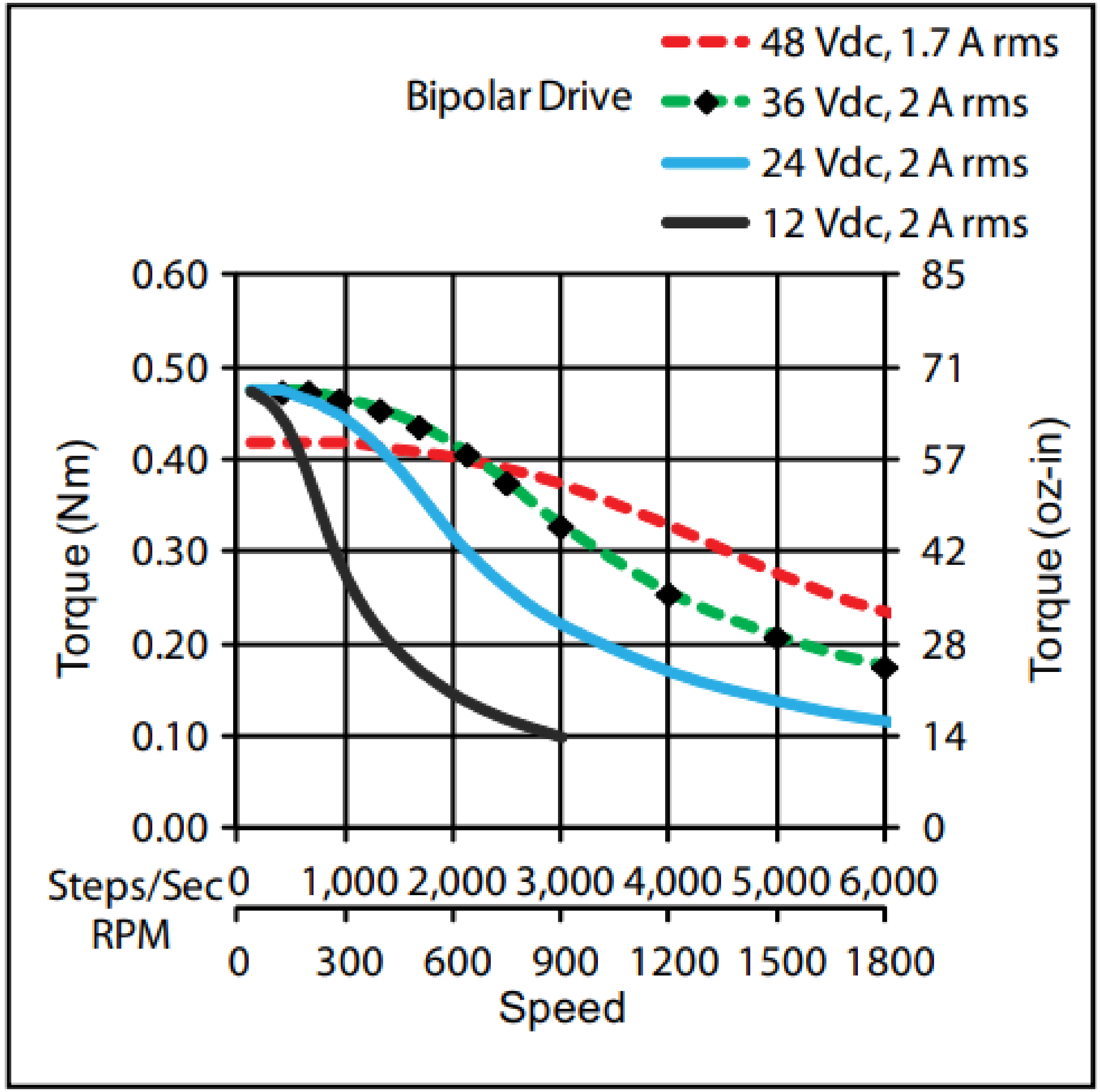
v	$(resolution * \frac{steps}{s})(GR)$	.225 m/s
Res.	$\frac{5mm}{200steps}$	.025mm
$\frac{steps}{s}$	Stepper motor @900rpm	3000 steps/s
GR	$\frac{Desired}{Max\ Motor\ rpm}$	3



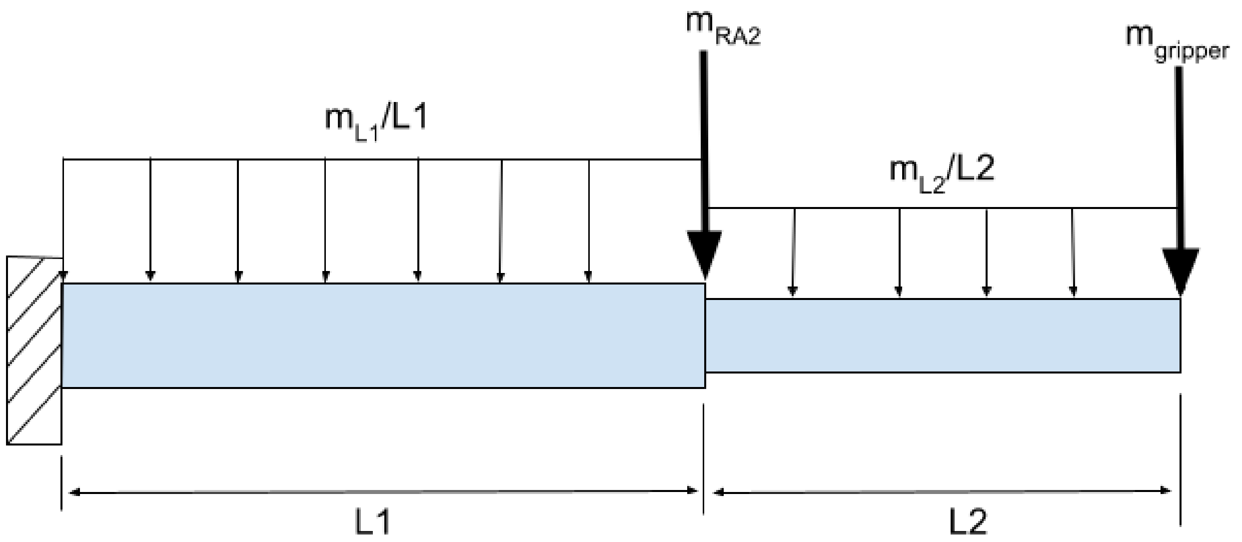


Linear Axis Information Graphs

Stepper Motor Torque Curve

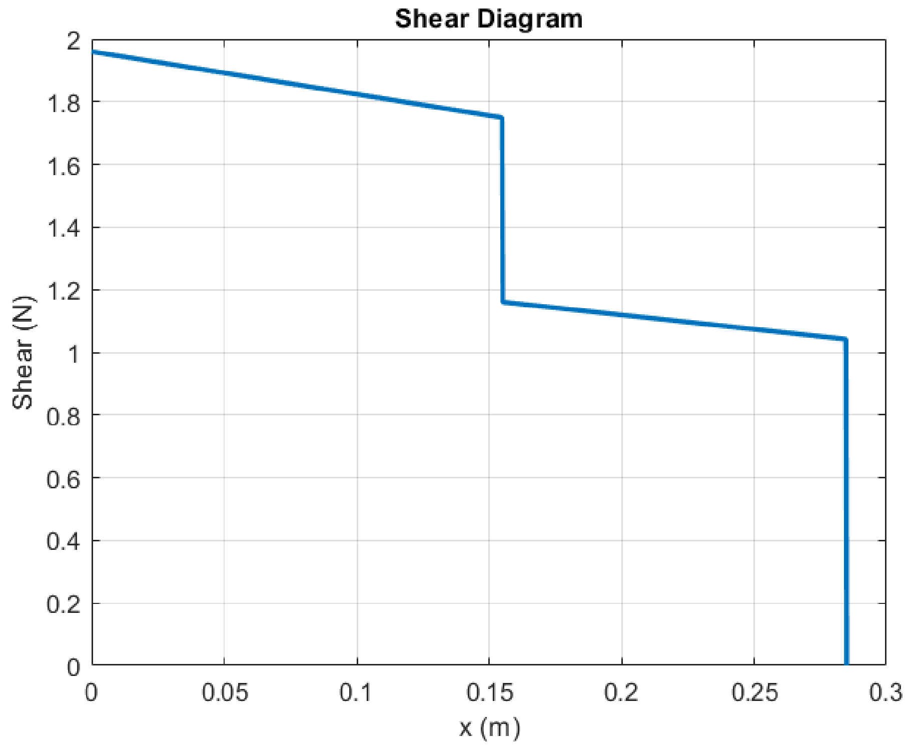


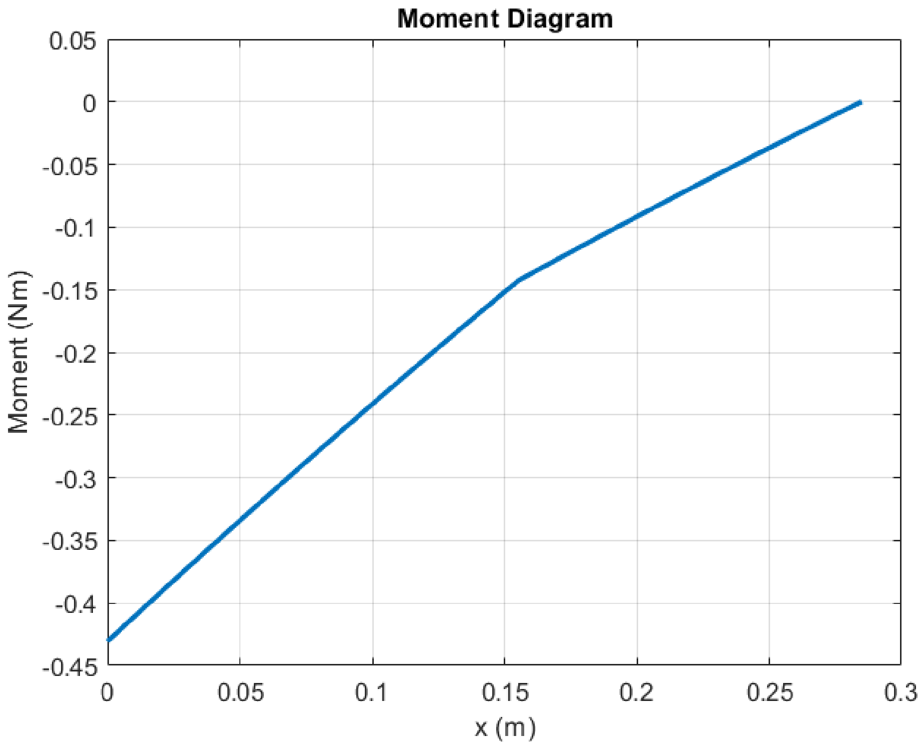
RA 1 Calculations - FBD



Variable	Description	Value
L1	Length of Link 1	15.5 cm
L2	Length of Link 2	13 cm
m_{L1}	Mass of Link 1	21.5 g
m_{L2}	Mass of Link 2	12.1 g
m_{RA2}	Mass of RA2	60 g
$m_{gripper}$	Mass of Gripper	103.2 g

RA 1 Calculations – Shear and Moment





Section (m)	Shear (N)	Moment (Ncm)
$0 \leq x \leq 0.155$	$V = 1.959 - 1.36x$	$M = 1.959x - 0.68x^2 - 0.43$
$0.155 \leq x \leq 0.285$	$V = 1.302 - 0.913x$	$M = 1.302x - 0.456x^2 - 0.334$

RA 1 Calculations – Bending Stress

- Cross-section [assume 2 links as 1]
  - a = 5 mm
  - b = 20 mm
  - $\sigma = \frac{Mc}{I}$

Variable (Inputs)	Definition	Value
M	Max Bending Moment	43.05 Ncm
c	Distance to neutral axis (b/2)	10 mm
I	Area Moment of Inertia ($ab^3/12$)	3333.3 mm^4
Variable (Outputs)	Definition	Value
σ	Bending Stress	1.29 MPa

RA 1 Calculations – Shear Stress

- $\tau_{max} = \frac{3V}{2A}$

Variable (Inputs)	Definition	Value
V	Max Shear Force	1.29 N
A	Cross Sectional Area	100 mm
Variable (Outputs)	Definition	Value
τ_{max}	Max Shear Stress	0.029 MPa

RA 1 Calculations – Torque Requirements

- $J_{total} = J_{L1} + J_{L2} + J_{RA2} + J_{gripper}$
- $J_{L1} = \frac{m_{L1} * b^2}{12} + \frac{m_{L1} * L_1^2}{3}$
- $J_{L2} = \frac{1}{12} m_{L2} * (w^2 + d^2) + m_{L2} \left(L_1 + \frac{L_2}{2}\right)^2$
- $J_{RA2} = m_{RA2} * L_1^2$
- $J_{gripper} = m_{gripper} * (L_1 + L_2)^2$

Variable (Inputs)	Definition	Value
m_{L1}	Mass of link 1	0.0215 kg
b	Height of link 1	20 mm
L_1	Length of link 1	15.5 cm
m_{L2}	Mass of link 2	0.0121 kg
w	Width of link 2	4 mm
d	Depth of link 2	20 mm
L_2	Length of link 2	13 cm
m_{RA2}	Mass of RA2	60 g
$m_{gripper}$	Mass of gripper	106.2 g
Variable (Outputs)	Definition	Value
J_{L1}	Inertia of Link 1 about RA1	$1.729 * 10^{-4}\ kg * m^2$
J_{L2}	Inertia of Link 2 about RA1	$5.861 * 10^{-4}\ kg * m^2$
J_{RA2}	Inertia of RA2 about RA1	$0.001\ kg * m^2$
$J_{gripper}$	Inertia of Gripper about RA1	$0.009\ kg * m^2$
J_{total}	Total Mass Moment of Inertia about RA1	$0.011\ kg * m^2$

RA 1 Calculations – Torque Requirements

- $L = \left(m_{L1} * \frac{L_1}{2} + m_{L2} * \left(L_1 + \frac{L_2}{2} \right) + m_{RA2} * L_1 + m_{gripper} * (L_1 + L_2) \right) * g$

Variable (Outputs)	Definition	Value
L	Gravitational Load	43.047 Ncm

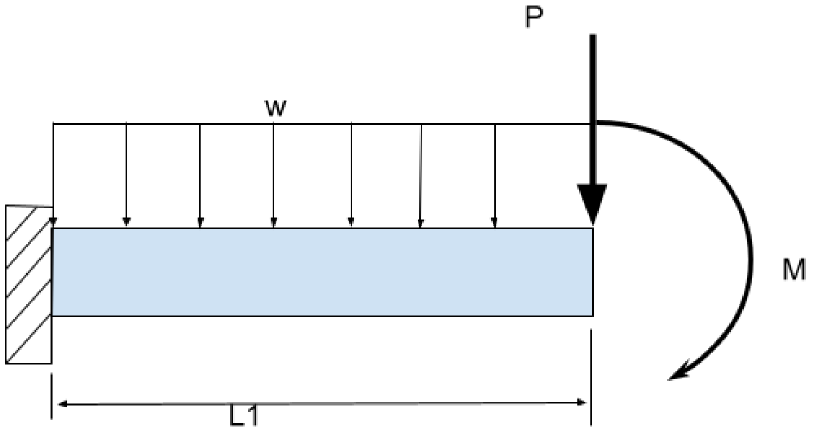
- $T = J_{total} * \alpha + L$

Variable (Inputs)	Definition	Value
J_{total}	Total Mass Moment of Inertia about RA1	$0.011\ kg * m^2$
α	Angular acceleration	$52.4 rad/s^2$
L	Gravitational Load	$43.047\ Ncm$
Variable (Output)	Definition	Value
T	Total Required Torque	$99.778\ Ncm$

RA 1 Calculations – Resolution

- Achievable Resolution with Servo Motor = 0.5°
- Total Link Length Actuated by RA1 = 28.5 cm
- End Effector Resolution = 28.5sin(0.5°) = 0.25 cm

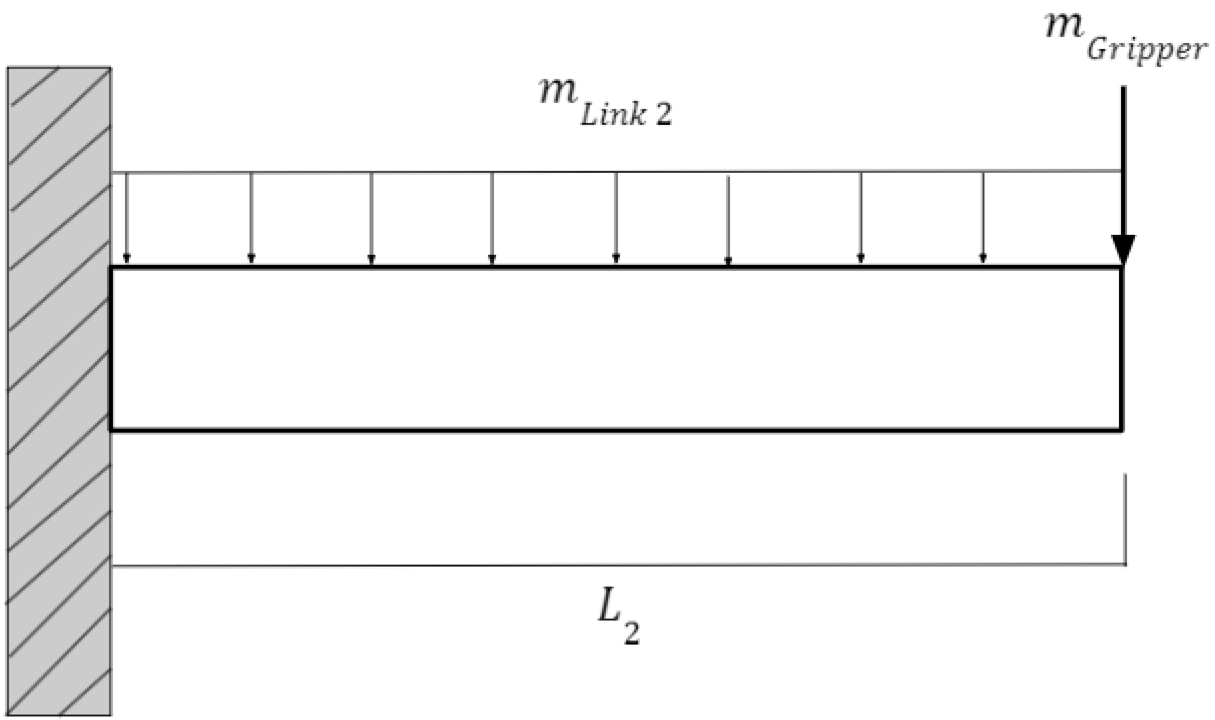
RA1 Calculations – End Deflection



$$v_{max} = \frac{-PL^3}{3EI} - \frac{ML^2}{2EI} - \frac{wL^4}{8EI}$$

Variable (Input)	Description	Value
w	Distributed Load of Link 1	1.36 N/m
P	Load of RA2, Link 2, and Gripper	1.749 N
M	Moment Caused by RA2, Link 2, and Gripper about RA1	0.414 Nm
L1	Length of Link 1	15.5 cm
E	Young's Modulus of PLA	3.5 GPa
I	Area Moment of Inertia	3333 mm^4
Variable (Output)	Description	Value
v	Deflection at the end of link 1	0.621 mm

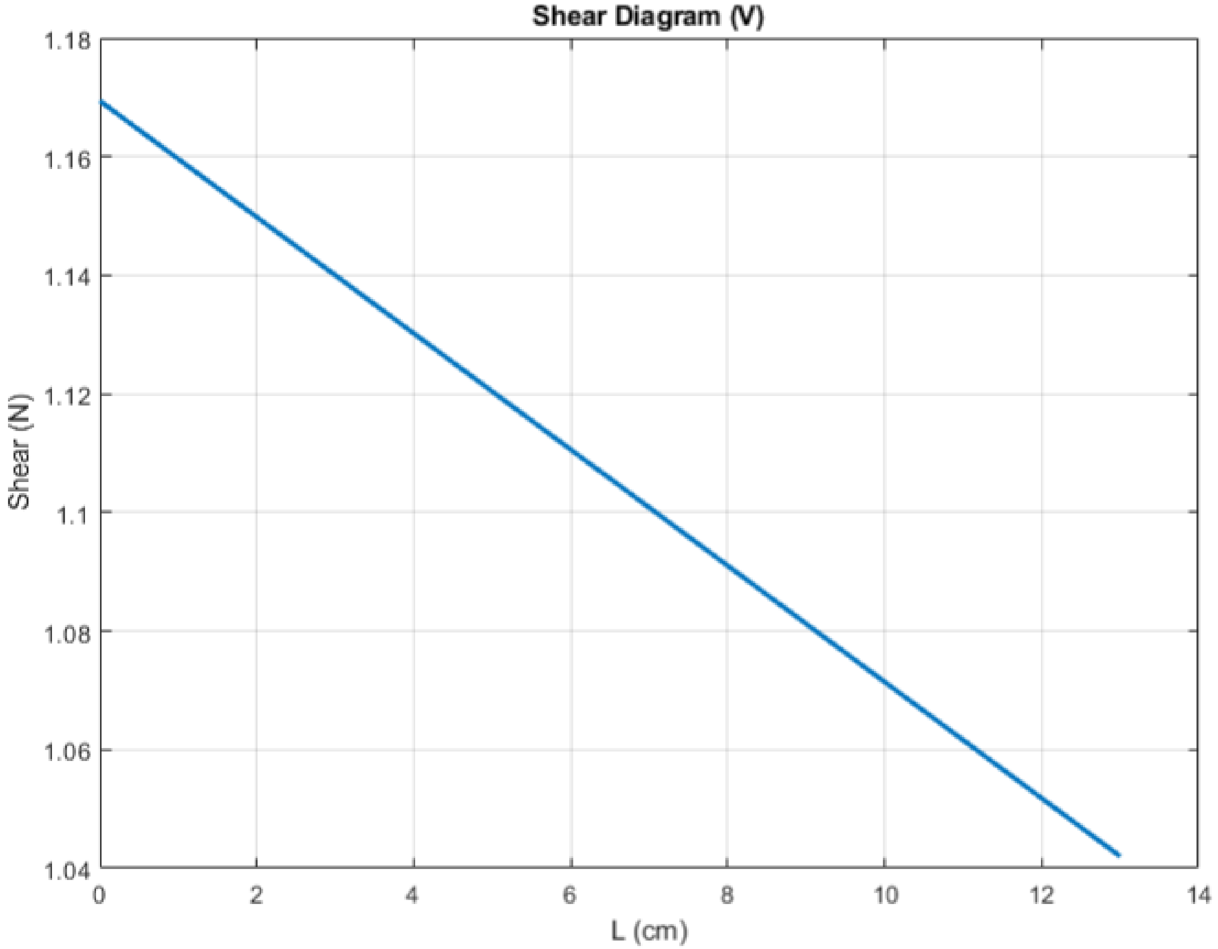
RA 2 Calculations: FBD

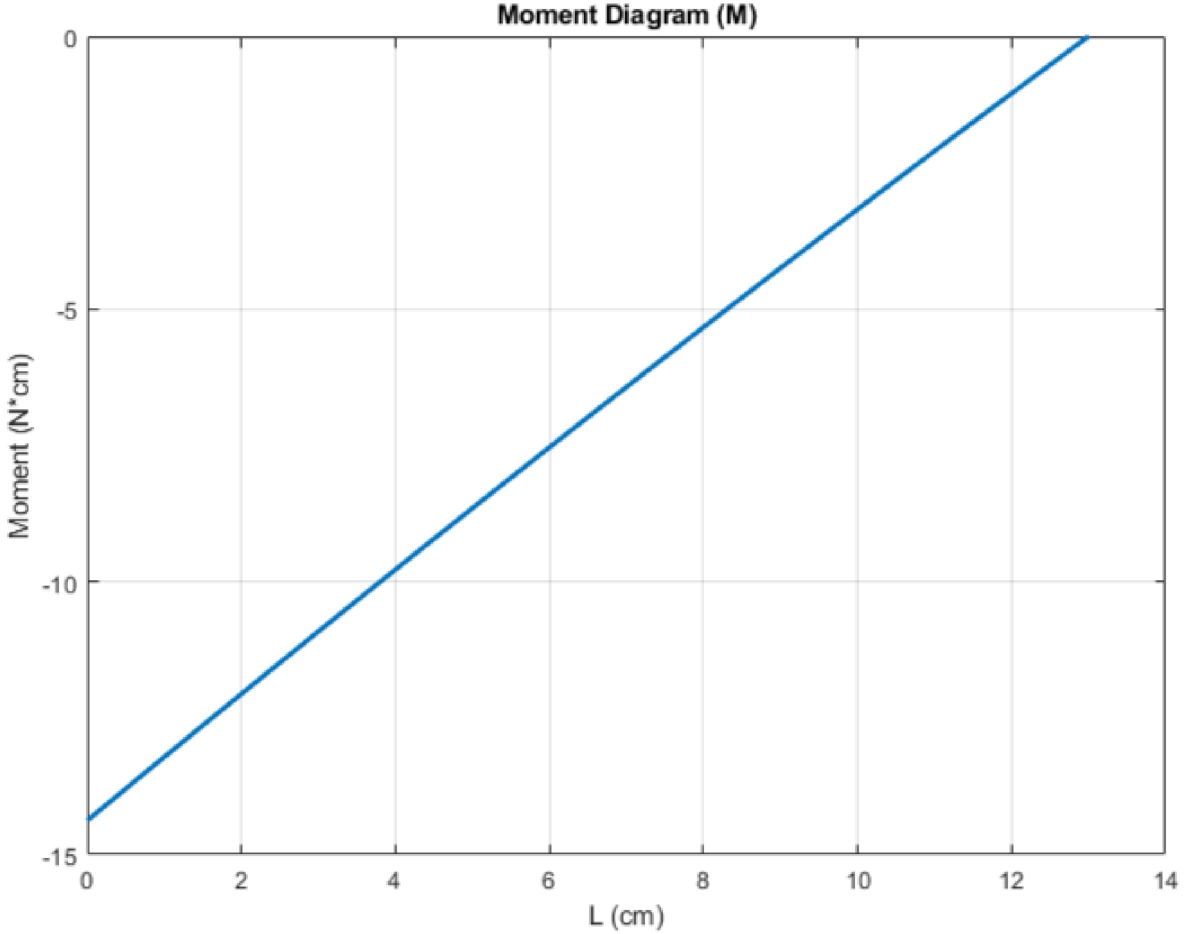


Variable	Description	Value
L2	Length of Link 2	13 cm
$m_{Link\ 2}$	Mass of Link 2	12.1 g
$m_{gripper}$	Mass of Gripper	106.2 g

Section (m)	Shear (N)	Moment (Ncm)
$0 \leq x \leq 13$	$1.1694 - 0.0098x$	$1.1694x - 0.0049x^2 - 14.3741$

RA 2 Calculations: Sheer/Moment Diagrams

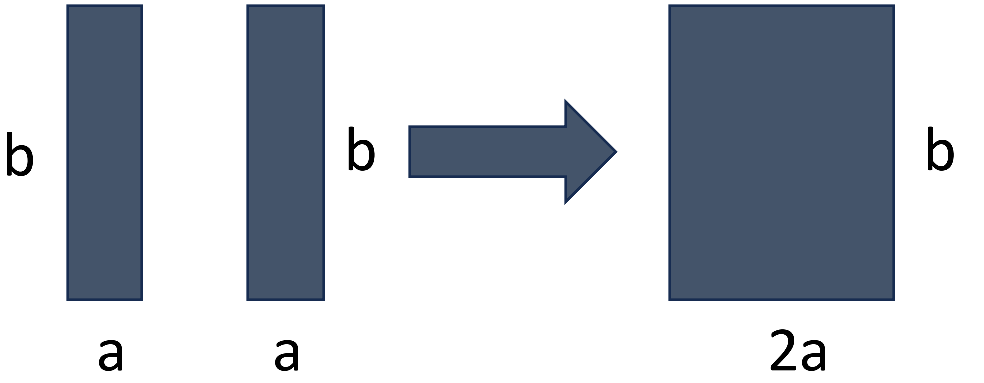




RA 2 Calculations: Bending Stress

Variable (Inputs)	Definition	Value
M	Max Bending Moment	14.3741 Nm
c	Distance to neutral axis (b/2)	10 mm
I	Area Moment of Inertia ($ab^3/12$)	2666.7 mm^4
Variable (Outputs)	Definition	Value
σ	Bending Stress	0.5390 MPa

$$\sigma = \frac{Mc}{I}$$



Cross section of two links combined for analysis
Where a = 2 mm and b = 20 mm

RA 2 Calculations: Shear Stress

Using the Equation: $\tau = \frac{3V}{2A}$

Variable (Inputs)	Definition	Value
V	Max Shear Force	1.1694 N
A	Cross Sectional Area	40 mm^2
Variable (Outputs)	**Definition**	**Value**
τ τ	Max Shear Stress	0.0219 MPa

RA 2 Calculations: Torque

Variable (Inputs)	Definition	Value
m_{L2}	Mass of link 2	12.1 g
m_g	Mass of gripper	106.2 g
b	Height of link 2 at cross section	20 mm
L_2	Length of link 2	130 mm
Variable (Outputs)	**Definition**	**Value**
J_z(Link 2)	Inertia of Link 2 about RA2	0.0018 mm^2
J_z(Point Mass)	Inertia of Gripper about RA2	7.33·10−7 mm^2
J_z(Total)	Total Mass Moment of Inertia about RA2	0.0018 mm^2

Using the Equation:

$$J_z(Total) = J_z(PointMass) + J_z(Link2)$$

Where:

$$J_z(Point\ Mass) = m_g + L_2^2$$

$$J_z(Link\ 2) = \frac{m_2 b_2^2}{12} + \frac{m_2 L_2^2}{3}$$

RA 2 Calculations: Torque Continued

$$L = \left(m_2 \frac{L_2}{2} + m_g L_2\right) g$$

Where the variables are the same as before and g is gravity

Variable (Outputs)	Definition	Value
L	Gravitational Load	14.37 Ncm

RA 2 Calculations: Torque Continued

Now Using the Equation: $T = J_z(Total) \cdot \alpha + L$

Variable (Inputs)	Definition	Value
Jz(Total)	Total Mass Moment of Inertia about RA1	0.0019 kgm2
α	Angular acceleration	52.4 radss2
L	Gravitational Load	0.1437 kgm
Variable (Output)	Definition	Value
T	Total Required Torque	24.16 Ncm

RA 2 Calculations: Resolution

- Achievable Resolution with Servo Motor = 0.5°
- Total Link Length Actuated by RA1 = 13 cm
- End Effector Resolution = 13sin(0.5°) = 0.113 cm

RA 2 Calculations: Deflection

Variable (Input)	Description	Value
w	Distributed Load of Link 2	0.98 N/m
P	Load of Gripper	1.0418 N
M	Moment Caused by Gripper about RA2	0.135 Nm
L2	Length of Link 2	13 cm
E	Young's Modulus of PLA	3.5 GPa
I	Area Moment of Inertia	2666.7 mm^4
Variable (Output)	Description	Value
v	Deflection at the end of link 2	0.0821 mm

Using the Equation:

$$v = -\frac{PL^3}{3EI} - \frac{ML^2}{2EI} - \frac{wL^2}{8EI}$$

Gripper Calculations – Required Torque (Constants)

- $\gamma = L * a$
- $F_{req} = \frac{m(2.3g+\gamma)}{\mu} FS$
- $F_1 = F_{req} \cos \alpha$
- $F_2 = \frac{F_1 \sin(\alpha+\beta)}{\sin \varphi}$
- $F_{gripper} = F_1 \cos(\alpha + \beta) + F_2 \cos \varphi$
- $T_{req} = F_{gripper} * r$

Variable (Inputs)	Definition	Value
L	Maximum Length of Robot	.3722m
a	Angular Acceleration (Robot)	$52.4 \frac{rads}{s^2}$
γ	Angular Acceleration (Cube)	$19.503 \frac{m}{s^2}$
m	Maass of cube	10g
g	Gravity Constant	$9.81 \frac{m}{s^2}$
μ	Coefficient of Friction	.63
FS	Factor of Safety	4
F_{req}	Force Required	2.6722
r	Radius of Rotation	.023m

Gripper Calculations – Required Torque (Min. Aperture)

- $\gamma = L * a$
- $F_{req} = \frac{m(2.3g+\gamma)}{\mu} FS$
- $F_1 = F_{req} \cos\alpha$
- $F_2 = \frac{F_1 \sin(\alpha+\beta)}{\sin\varphi}$
- $F_{gripper} = F_1 \cos(\alpha+\beta) + F_2 \cos\varphi$
- $T_{req} = F_{gripper} * r$

Variable (Inputs @ minimum aperture)	Definition	Value
α	Angle	60.46°
β	Angle	7.54°
φ	Angle	82.46°
F_1	Force on Gear to Link joint	1.317N
F_2	Force on Link to Effector joint	1.233N
$F_{gripper}$	Force normal to Effector	.6550N

Variable (Output)	Definition	Value
T_{req}	Total Required Torque (minimum aperture)	.01507Nm

Gripper Calculations – Required Torque (Max. Aperture)

- $\gamma = L * a$
- $F_{req} = \frac{m(2.3g+\gamma)}{\mu} FS$
- $F_1 = F_{req} \cos\alpha$
- $F_2 = \frac{F_1 \sin(\alpha+\beta)}{\sin\varphi}$
- $F_{gripper} = F_1 \cos(\alpha + \beta) + F_2 \cos\varphi$
- $T_{req} = F_{gripper} * r$

Variable (Inputs @ maximum)	Definition	Value
α	Angle	22.05°
β	Angle	67.95°
φ	Angle	0°
F_1	Force on Gear to Link joint	2.4767
F_2	Force on Link to Effector joint	0N
$F_{gripper}$	Force normal to Effector	2.4767N

Variable (Output)	Definition	Value
T_{req}	Total Required Torque (maximum aperture)	.0569Nm